
Training-Free Cultural Alignment of Large Language Models via Persona Disagreement

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Abstract

1 Large language models are increasingly deployed in decisions that turn on moral
2 judgement, from content moderation to clinical prioritisation, yet they answer as
3 if the whole world thinks like the West. The Moral Machine experiment [Awad
4 et al., 2018] showed this is wrong at scale: 40 million judgments across 233 coun-
5 tries reveal that moral preferences are systematically structured by culture, and a
6 model that ignores this variation does not merely underperform. It imposes one
7 society’s intuitions on all others. Existing fixes do not scale to global deployment:
8 fine-tuning needs per-country preference data and GPU budgets, reward-guided
9 decoding needs per-country reward models, and activation steering needs access
10 to model internals that black-box APIs do not expose. We work in the only regime
11 that scales: inference time, with no weight updates, no training data, and no inter-
12 nal access. The key observation is that within-country demographic *disagreement*,
13 not consensus, is the steering signal: when culturally grounded personas agree, the
14 base model is already calibrated; when they disagree, the spread tells us what to fix
15 and by how much. DISCA (Disagreement-Informed Steering for Cultural Align-
16 ment) instantiates each country as a panel of four World-Values-Survey-grounded
17 persona agents [Haerpfner et al., 2022], converts their disagreement into a bounded,
18 loss-averse correction whose magnitude is set by the panel’s variance, and shrinks
19 the correction toward zero when the estimate is unreliable. Across 20 countries
20 and 7 open-weight backbones (2B–70B) from five model families, DISCA re-
21 duces cultural misalignment on MultiTP [Jin et al., 2025] by 10–24% on binary
22 moral dilemmas and 2–7% on open-ended scenarios; a 14B backbone with DISCA
23 reaches lower absolute misalignment than a vanilla 70B model.

24 1 Introduction

25 Large language models increasingly mediate decisions that turn on moral judgement: content mod-
26 eration policies, clinical prioritisation guidelines, autonomous-vehicle behaviour specifications, and
27 high-stakes recommendation pipelines. In each of these settings, the model’s implicit moral pref-
28 erences become the system’s moral preferences. Yet a growing body of evidence shows that these
29 implicit preferences are not culturally neutral. The Moral Machine experiment collected 40 million
30 judgments across 233 countries and showed that moral preferences are systematically structured by
31 culture [Awad et al., 2018]; LLM outputs correlate most strongly with respondents from Western,
32 educated, industrialised, rich, and democratic populations [Henrich et al., 2010, Santurkar et al.,
33 2023]; and post-training alignment introduces further unintended cultural shifts [Ryan et al., 2024,
34 Zewail et al., 2026]. A model that ignores this variation does not merely underperform on a bench-
35 mark; it imposes one society’s intuitions on every user it serves.

36 The cultural gap is now well-documented at the model level. The MultiTP benchmark [Jin et al.,
37 2025] extends the Moral Machine protocol to 107 languages with country-specific Average Marginal

Component Effects (AMCEs), and reports that none of 19 evaluated LLMs recovers the country-level structure of human moral preferences. The asymmetries are substantial: Japan ranks first among countries on preference for sparing pedestrians, China ranks 116th, and current models do not capture either extreme. Concurrent work shows that sociodemographic persona prompts shift LLM moral decisions two to four times more than they shift human decisions [Kim et al., 2025], indicating that the models are not insensitive to cultural framing, but that the framing they receive at deployment time does not match the cultural distribution they are asked to serve.

Closing this gap is harder than it looks. Three families of methods have been proposed, and each demands resources that do not scale to the long tail of countries served by a single deployed model. Fine-tuning and culture-aware adapters require per-country preference datasets and GPU budgets for every target population [Zhang et al., 2026, Yao et al., 2025]; reward-guided decoding requires a separate trained reward model per country [Khanov et al., 2024, Mudgal et al., 2024]; and activation steering requires write access to model internals [Arditi et al., 2025, Zou et al., 2023, Turner et al., 2024], which black-box APIs do not expose. Any deployable solution must therefore work at inference time, must not assume per-country reward models or labelled preference data, and must not assume access to model internals beyond what an API exposes. We refer to this as the black-box, public-data-only regime, and argue that it is the only regime in which cultural alignment can scale from a handful of curated countries to the hundreds of cultures a globally-deployed model actually serves.

We make the following observation. When certain culturally grounded persona prompts representing the same country face the same moral dilemma, the dispersion of their decoding-time logit gaps is itself informative. If they agree, the base model is already calibrated for that country and no correction is needed; if they disagree, the spread of their gaps encodes both the direction and the magnitude of the needed correction. Our method, DISCA (Disagreement-Informed Steering for Cultural Alignment), instantiates each country as a panel of four persona agents grounded in World Values Survey microdata [Haerpfer et al., 2022], evaluates them in a single batched forward pass, and converts the panel’s variance into a bounded, loss-averse logit correction whose magnitude is automatically attenuated when the underlying estimate is unreliable. We show that this attenuation is not heuristic: it implements a variance-aware shrinkage estimator whose shrinkage weight is determined by the panel’s empirical disagreement, formally characterising when the correction can be trusted.

Across 20 countries spanning four continents and seven open-weight backbones (2B–70B parameters) from five model families, DISCA reduces cultural misalignment on MultiTP by 10–24% on binary moral dilemmas and 2–7% on open-ended scenarios. The strongest absolute alignment is achieved by Phi-4 (14B), which surpasses a vanilla Llama-3.3-70B backbone despite using one fifth of the parameters; this suggests that, in the regime we study, calibration competes with scale rather than merely complementing it. An evaluation on the BLEnD factual cultural QA benchmark [Myung et al., 2024] reveals a clean scope boundary: DISCA steers values, where the decision reduces to a scalar logit gap, but does not transfer to factual recall, where the decision is a single token in a large vocabulary. Failure modes are architecturally rather than culturally determined and are diagnosable from the base model’s vanilla logit conditioning alone.

Contributions.

- An inference-time cultural alignment method that requires no weight updates, no per-country reward models, and no white-box access.
- A formal characterisation of the disagreement-driven correction as a variance-aware shrinkage estimator, where panel variance determines the shrinkage weight and a dual-pass reliability gate provides a finite-sample plug-in estimator of this weight.
- An empirical evaluation across 20 countries, four continents, seven open-weight backbones, and three evaluation formats (binary moral dilemmas, open-ended scenarios, factual QA), demonstrating that a 14B model with the proposed recipe matches or surpasses a 70B model without it.

89 2 Related Work

90 **Benchmarks and cultural evaluation.** The Moral Machine experiment [Awad et al., 2018] es-
 91 tablished that moral preferences are culturally structured; MultiTP [Jin et al., 2025] extends this to
 92 107 languages with country-specific AMCEs. Takemoto [2024] and Ahmad and Takemoto [2025]
 93 showed that LLM alignment varies widely across model families. Kim et al. [2025] demonstrated
 94 that sociodemographic personas induce moral-decision shifts in LLMs $2\text{--}4\times$ larger than in humans,
 95 with political orientation producing the largest effect (a “partisan sorting” phenomenon absent in
 96 human respondents). Their finding that simple persona prompts amplify rather than correct mis-
 97 alignment directly motivates DISCA’s approach: we use WVS-grounded personas not to steer de-
 98 cisions toward a single viewpoint but to measure within-country disagreement and convert it into
 99 a bounded correction. Existing moral benchmarks remain English-centric [Hendrycks et al., 2021,
 100 Jin et al., 2022, Nie et al., 2023] or cover fewer languages [Kirk et al., 2024, Durmus et al., 2023,
 101 Qiu et al., 2025]. We validate on MultiTP’s binary trolley vignettes, the BLEnD factual cultural
 102 QA benchmark [Myung et al., 2024] (Appendix A6), and a dedicated open-ended moral-dilemma
 103 track (§5). On the diagnostic side, LLM outputs correlate most with WEIRD respondents [Santurkar
 104 et al., 2023], RLHF introduces unintended cultural shifts [Ryan et al., 2024], and recent work shows
 105 LLMs systematically stereotype non-Western moral values [Zewail et al., 2026].

106 **Inference-time alignment.** The pluralistic alignment vision [Sorensen et al., 2024] argues that
 107 inference-time methods are a natural vehicle for cultural adaptation, but existing approaches each
 108 sacrifice one of the three constraints we target: activation steering [Arditi et al., 2025, Zou et al.,
 109 2023] requires white-box access; reward-guided decoding [Khanov et al., 2024, Mudgal et al.,
 110 2024] needs per-country reward models; culture-aware adapters [Zhang et al., 2026, Yao et al.,
 111 2025] need per-culture training. Recent logit-level steering [An et al., 2026, Khanuja et al., 2026]
 112 demonstrates that the logit surface is a richer control interface than appreciated, but neither targets
 113 country-conditioned moral calibration. Black-box alignment modules [Bobbili et al., 2025, Chen
 114 et al., 2025] still require a training stage, and adaptive IS in pre-logit space [Kanai et al., 2025] re-
 115 quires activation access. Training-time prospect-theoretic alignment [Ethayarajh et al., 2024] applies
 116 the same PT value shape we use, but at training time; DISCA applies it at *test time*, per scenario,
 117 aggregating heterogeneous personas rather than shaping a trained policy.

118 **Distinction from superficially similar paradigms.** DISCA may superficially resemble ensemble
 119 methods, but the resemblance is architectural, not algorithmic. In self-consistency [Wang et al.,
 120 2023] and LLM debate [Du et al., 2024, Liang et al., 2024], diversity is noise to be eliminated via
 121 majority vote; in standard ensembles, multiple outputs are averaged to reduce variance of the *predic-*
 122 *tion*. In DISCA, diversity *is* the signal: the *spread* of persona logit gaps, not the mode, determines
 123 the correction magnitude. This distinction is not merely conceptual. We prove (Theorem 1) that
 124 the disagreement-driven correction is a James–Stein shrinkage estimator [James and Stein, 1961]
 125 that strictly dominates unshrunk consensus in worst-case MSE: ensembles average to reduce output
 126 variance, whereas DISCA uses within-panel variance as a sufficient statistic to *control* correction
 127 magnitude. Standard calibration (temperature scaling, MC-Dropout [Gal and Ghahramani, 2016,
 128 Kwon et al., 2026]) adjusts confidence isotropically; DISCA applies anisotropic, loss-averse correc-
 129 tions. The ablation (Table 5) confirms the distinction empirically: removing personas degrades MIS
 130 by $+0.047\text{--}0.061$, while removing the reliability gate loses $+0.015\text{--}0.021$, effects no ensemble or
 131 calibration method reproduces.

132 To our knowledge, DISCA is the first training-free method targeting country-level moral calibra-
 133 tion. The method is grounded in control-as-inference, Prospect Theory, and importance-sampling
 134 variance control (Appendix A12).

135 3 Method: DISCA

136 DISCA produces a small adjustment to the model’s decision-token logits, computed from a number
 137 of culturally grounded personas and applied at decode time. The size of the adjustment is not fixed:
 138 when these personas agree on what the country’s preference should be, we apply the adjustment in
 139 full; when they disagree, we apply only a fraction of it, scaled by how much the panel disagrees.
 140 The rest of this section makes this idea precise. We first fix notation in 3.1, construct the culturally

grounded panel in 3.2, describe batched evaluation and debiasing in 3.3 and 3.4, and then formalise the disagreement-driven correction in 3.5.

3.1 Setup

We consider a frozen language model f_θ deployed on a forced-choice task: each input x asks the model to choose between two options via decision tokens A and B . Let $z(x) = [z_a, z_b]$ denote the model’s logits at the decision-token position, and let $\delta(x) = z_b - z_a$ be the resulting decision gap. For target country c , we assume access only to f_θ as a black box and to a public characterisation of country c ’s population (in our case, World Values Survey microdata [Haerpfer et al., 2022]); we do not assume per-country preference labels, reward models, or write access to θ .

Our goal is to compute, from f_θ alone, a per-scenario correction $\delta^*(x, c)$ such that the corrected logit gap $\delta(x) + \delta^*(x, c)$ produces decisions better aligned with country c ’s preferences than the base gap $\delta(x)$ alone. We measure alignment in ℓ_2 distance between the model’s preference vector and a country-level reference vector; the benchmark, the reference vector, and the precise alignment metric are described in §4. The remainder of §3 focuses on how δ^* should depend on the panel, what formal property this dependence has, and how to estimate the correction from finite samples. The next subsection states the principle that governs the magnitude of δ^* .

3.2 Cultural Persona Construction

We use $N=4$ personas (three age cohorts plus a country-wide aggregate) grounded in WVS Wave 7 microdata [Haerpfer et al., 2022]. Four is the smallest panel covering the three demographic axes with most within-country variance (young, middle-aged, older) while preserving a population anchor [Inglehart and Welzel, 2005, Kish, 1965]. Ablation confirms the panel is load-bearing: removing it costs $+0.083$ MIS, the largest single-component drop (Table 5). A sensitivity check replacing the aggregate persona with a utilitarian anchor yields marginal $\Delta = -0.02$ MIS (Appendix A15.4).

For each country c , following the 10-variable cultural-value scheme of Greco et al. [2026], we extract: *religiosity* (Q6P), *child-rearing values* (Q17P), *moral acceptability* (Q177–Q182), *social trust* (Q57P), *political participation* (Q199P, Q200P), *national pride* (Q254P), *happiness* (Q46P), *gender equality* (Q29P–Q33P), *materialism orientation* (Q152–Q153), and *tolerance for diversity* (Q19P–Q23P). Items from the “inverted” WVS-7 CSV (suffix P) are directionally coded so that higher values consistently indicate the positive pole of each dimension. For each country, respondents are segmented into three age cohorts—*young* (<36), *middle-aged* (36–55), and *older* (>55)—yielding three cohort-specific personas that capture generational variation in cultural values. A fourth persona aggregates all respondents regardless of age, anchoring the ensemble with the country’s population-wide cultural centre of gravity. All four personas are thus data-driven and country-specific; no agent carries a hard-coded moral stance that could bias particular dimensions. Dimension means are normalised to $[0, 1]$ and mapped to four-level natural-language descriptors via quartile cuts (≥ 0.75 , ≥ 0.50 , ≥ 0.25 , < 0.25); see Appendix A15. Each persona is a native-language system prompt assembled from a header, one sentence per WVS dimension with data, and a moral-reasoning closing line.

WVS grounding ensures empirically measured within-country variation rather than stereotypes [Deshpande et al., 2023]. The 10-dimensional WVS profile explains $R^2 \in [0.55, 0.69]$ of human AMCE variance across the 20-country panel (Appendix A15.3), and a leave-one-dimension-out probe identifies load-bearing couplings alongside noisy ones. Crucially, the IS stage automatically down-weights persona signals that increase collective disutility, acting as a built-in noise filter (Appendix A15.3). A WVS dropout study confirms a concentrated importance profile: religiosity, gender equality, and moral permissiveness are load-bearing ($\Delta\text{MIS} > 0.03$), while national pride and life satisfaction have near-zero impact (Appendix A7.5). Persona examples are in Appendix A15.

3.3 Batched Evaluation and Logit Extraction

All $N + 1 = 5$ input sequences (base + 4 personas) are evaluated in a *single* batched forward pass:

$$\mathbf{z}_{\text{base}}, \mathbf{z}_1, \dots, \mathbf{z}_N = f_\theta(\mathbf{x}; \emptyset, \mathbf{s}_1, \dots, \mathbf{s}_N). \quad (1)$$

Per-category logit temperatures T_{cat} scale logits before computing decision gaps. We use $T_{\text{cat}} = 4.0$ (Species), 3.5 (Gender), and 1.5 for Age, Fitness, Social Value, and Utilitarianism (Table 1); the same values appear in Appendix A10.

$$\delta_i = \frac{z_{i,b} - z_{i,a}}{T_{\text{cat}}}, \quad \delta_{\text{base}} = \frac{z_{\text{base},b} - z_{\text{base},a}}{T_{\text{cat}}}, \quad \bar{\delta} = \frac{1}{N} \sum_{i=1}^N \delta_i \quad (\text{consensus}), \quad (2)$$

$$r_i = \delta_i - \delta_{\text{base}} \quad (\text{per-agent reward}). \quad (3)$$

Here δ_{base} is the base-prompt decision gap; r_i measures how strongly persona i shifts the model away from the neutral baseline. (The scalar r in Eq. 5 denotes the bootstrap reliability weight, not a reward.)

3.4 Positional Debiasing

To cancel recency bias [Liu et al., 2024], each scenario is evaluated twice: once in the original ordering, once with *both* choice labels ($A \leftrightarrow B$) and group labels (Group A $\leftrightarrow$ Group B) swapped in the native language. The debiased decision gap for each agent is $\delta_i = (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$; the base prompt yields δ_{base} under the same rule. Implementation details (placeholder-based three-step swap to avoid double-substitution bugs in native scripts) are in Appendix A14.5.

3.5 From Persona Disagreement to a Bounded Correction

The persona consensus $\bar{\delta}$ already captures the cultural direction, but applying it directly would ignore two problems: the consensus may overcorrect toward one loud persona, and finite-sample noise may push the correction in an unreliable direction. We address both through importance sampling with a cooperative, loss-averse utility.

We sample candidate perturbations $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ around the consensus and score each by how much it improves alignment with every persona, not just the loudest one. Per-persona gains are aggregated through a Prospect-Theory value function [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] with $\kappa \approx 2.25$, encoding a “do no harm” prior: the PT kernel is a *controller design choice*, an asymmetric aggregation kernel where harming one demographic group is penalised more than equally-sized help to another, not a literal model of LLM cognition. Auxiliary runs with linear and WVS-weighted kernels underperform the PT kernel, and robustness sweeps keep all JSD spans within ± 0.004 (Appendix A13). The softmax-weighted IS update can be read as approximate control-as-inference [Williams et al., 2018, Levine, 2018].

$$U_{\text{total}}(\epsilon_k) = (1 - \lambda_{\text{coop}}) \frac{1}{N} \sum_{i=1}^N v\left(\frac{g_{i,k}}{\sigma}\right) + \lambda_{\text{coop}} v\left(\frac{g_{\text{cons},k}}{\sigma}\right), \quad (4)$$

where $v(\cdot)$ is the Kahneman–Tversky value function with gain curvature $\alpha=0.88$ and loss aversion $\kappa=2.25$ [Kahneman and Tversky, 1979], $g_{i,k}$ measures how much candidate k improves over the base for persona i (Appendix A14), and $\lambda_{\text{coop}}=0.7$ weights the group consensus. Candidates are then softmax-weighted by utility and combined into a single correction δ_m^* ; an effective-sample-size guard zeros the correction when the weights collapse, falling back on the raw consensus (Appendix A14).

Sampling noise is handled by a *dual-pass reliability gate*: DISCA splits the IS budget into two independent passes and compares them:

$$\delta_{\text{IS}}^* = \underbrace{\exp(-(\delta_1^* - \delta_2^*)^2/s)}_{r \in (0,1]: \text{reliability weight}} \cdot \frac{\delta_1^* + \delta_2^*}{2}. \quad (5)$$

When the two passes concur, $r \approx 1$ and the full correction applies; when they diverge, r decays exponentially, requiring no hard threshold or manual tuning. This self-limiting property prevents confident-but-wrong corrections on noisy logit landscapes and provides the safety margin the open-ended extension (§3.6) relies on. The dual-pass split yields a bootstrap-style variance probe [Efron, 1979, Elvira and Martino, 2021] that flags instability rather than hiding it behind more samples, with an ESS guard [Kish, 1965] zeroing the offset when weights collapse. We treat the gate as a variance heuristic, not a formal unbiasedness certificate (Appendices A13–A14). The final correction

Algorithm 1 DISCA Inference: Scenario $\mathbf{x}$ in Country c

Require: Frozen LLM f_θ ; personas $\{\mathbf{s}_i\}_{i=1}^N$; hyperparameters $K_{\text{half}}, \sigma, \lambda, \eta, T_{\text{dec}}, s$

Ensure: $p_{\text{spare}} \in [0, 1]$

```

1: for order  $\in \{\text{orig}, \text{swap}\}$  do
2:   Batched forward:  $f_\theta$  on base +  $N$  personas
3:   Extract logit pairs; compute  $\delta_i^{(\text{ord})}$ 
4: end for
5:  $\delta_i \leftarrow (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$ 
6: Compute  $\bar{\delta}, r_i$  [Eqs. 2–3]
7: for  $m \in \{1, 2\}$  do
8:   for  $k = 1$  to  $K_{\text{half}}$  do
9:     Sample  $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ ;  $U_{\text{tot}}(\epsilon_k)$ 
10:   end for
11:    $\delta_m^* \leftarrow \sum_k w_k^{(m)} \epsilon_k$  (if ESS  $> \rho_{\text{eff}}$ )
12: end for
13:  $r \leftarrow \exp(-(\delta_1^* - \delta_2^*)^2/s)$ 
14:  $\delta_{\text{IS}}^* \leftarrow r(\delta_1^* + \delta_2^*)/2$ 
15:  $\delta_{\text{final}} \leftarrow \text{blend}(\bar{\delta}, \delta_{\text{base}}, \delta^*)$ 
16: return  $p_{\text{spare}} = \sigma(\delta_{\text{final}}/T_{\text{dec}})$ 

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231 combines the reliability-gated IS offset with an interpolation between persona consensus and the
 232 debiased base (Appendix A14).

233 This self-regulating behaviour has a precise formal justification.

234 **Theorem 1** (Optimality of disagreement-driven shrinkage). *Let $\delta_1, \dots, \delta_N$ be the persona logit*
 235 *gaps for a single scenario, modelled as $\delta_i = \delta_h + \eta_i$ with η_i i.i.d., $\mathbb{E}[\eta_i] = 0$, $\text{Var}(\eta_i) = \tau^2$, and*
 236 *δ_h the unknown human-preferred gap. Define the consensus correction $\Delta = \bar{\delta} - \delta_{\text{base}}$ and the*
 237 *within-panel disagreement $D^2 = \frac{1}{N-1} \sum_i (\delta_i - \bar{\delta})^2$. Then:*

238 (i) **Disagreement is a sufficient statistic for correction risk.** *The MSE decomposes as $\mathbb{E}[(\Delta -$*
 239 *$\Delta_h)^2] = \bar{b}^2 + \tau^2/N$, where $\bar{b}$ is the mean persona bias and D^2 is an unbiased estimator of*
 240 *τ^2 .*

241 (ii) **Shrinkage dominates consensus.** *For $N \geq 4$, the James–Stein shrinkage estimator $\hat{\Delta}_{\text{JS}} =$*
 242 *$\max(0, 1 - \frac{(N-3)D^2}{N\Delta^2}) \cdot \Delta$ strictly dominates the unshrunk consensus Δ in MSE uniformly*
 243 *over δ_h .*

244 (iii) **The reliability gate implements asymptotically optimal shrinkage.** *The dual-pass weight*
 245 *$r = \exp(-(\delta_1^* - \delta_2^*)^2/s)$ is a monotone decreasing function of the inter-replicate variance*
 246 *$V_r = (\delta_1^* - \delta_2^*)^2/2$, itself an unbiased estimator of τ^2/K_{half} . As $K_{\text{half}} \rightarrow \infty$, $r \cdot \Delta \rightarrow \hat{\Delta}_{\text{JS}}$*
 247 *in probability.*

248 *Proof sketch.* Part (i) follows from the bias-variance decomposition and the unbiasedness of the
 249 sample variance. Part (ii) is a direct application of the James–Stein theorem [James and Stein,
 250 1961]: for $N \geq 4$, shrinking a d -dimensional normal mean toward the origin dominates the MLE,
 251 and the persona model reduces to this setting with $d = N - 1 \geq 3$. Part (iii) follows because V_r is an
 252 unbiased estimator of τ^2/K_{half} (the two half-budget passes are independent), and the exponential
 253 map $r = e^{-V_r/s}$ is a consistent plug-in for the oracle shrinkage factor $\gamma^* = \Delta_h^2/(\Delta_h^2 + \tau^2/N)$
 254 under standard regularity (full proof in Appendix A1).

255 **Corollary 2.** *$N = 4$ is the minimum panel size guaranteeing that the disagreement-driven correc-*
 256 *tion dominates naive consensus in worst-case MSE. DISCA is therefore not an ensemble (which tar-*
 257 *gets the mean) but a shrinkage estimator that uses within-group disagreement to control correction*
 258 *magnitude, recovering an instance of the James–Stein phenomenon for inference-time alignment.*

259 The final spare probability is $p_{\text{spare}} = \sigma(\delta_{\text{final}}/T_{\text{dec}})$ with $T_{\text{dec}}=0.5$. Algorithm 1 summarises the
 260 full procedure; numerical values for all symbols appear in Table 20 (Appendix A10).

Table 1: The six moral dimensions in MultiTP. T_c is the per-attribute logit temperature used by our method; values reflect empirical logit magnitudes and are validated in App. A10.

Dimension	Preferred	Contra	T_c
Species	Human	Animal	4.0
Gender	Female	Male	3.5
Age	Young	Elderly	1.5
Fitness	Fit	Unfit	1.5
Social Value	High (exec.)	Low (homeless)	1.5
Utilitarianism	More lives	Fewer lives	1.5

3.6 Extension to Open-Ended Ethical Scenarios

To extend DISCA to open-ended scenarios, the model generates free-form responses under both the original prompt and four persona-specific prompts. An LLM then parses these outputs to return a {choice, confidence} pair. The continuous decision signal is represented by a pseudo-logit-gap (the A/B logit distance) extracted by the LLM. The dual-pass PT-IS mechanism is applied similarly to the binary case.

4 Experimental Setup

4.1 Dataset and metric

We evaluate on MultiTP [Jin et al., 2025], which adapts the Moral Machine trolley-problem protocol [Awad et al., 2018] to language-model evaluation. Each scenario x describes two groups of people the autonomous vehicle could spare and asks the model to choose between them via decision tokens A and B . Each scenario isolates one of six moral attributes (Table 1), with the two groups matched on the other five. As a concrete example, an Age scenario might ask the model whether to spare a group of three young children or three elderly adults; across many such Age scenarios, the average rate at which the model spares the younger group becomes the model’s AMCE on Age.

Following the Moral Machine convention, one side of each attribute is labelled the *preferred* side. The label is a measurement convention from conjoint analysis [Awad et al., 2018], not a normative claim: it names the side the AMCE measures *toward*, so that AMCE values are signed consistently across attributes. We assign decision token B to the preferred group and token A to the contra group, partitioning the country- c scenario pool into six attribute-specific groups $\mathcal{S}_d^{(c)}$.

Sparing probability and AMCE. The frozen model f_θ produces logits $z(x) = [z_a, z_b]$ at the decision-token position; the decision gap is $\delta(x) = z_b - z_a$. The sparing probability under decoding temperature T_{dec} is $p_{\text{spare}}(x) = \sigma(\delta(x)/T_{\text{dec}})$, the model’s probability of sparing the preferred group on scenario x . The model’s Average Marginal Component Effect (AMCE) on attribute d is the empirical mean of p_{spare} across $\mathcal{S}_d^{(c)}$:

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{x \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(x). \quad (6)$$

Stacking across the six attributes gives the model’s AMCE vector $\hat{m}^{(c)} \in \mathbb{R}^6$. The human AMCE vector $h^{(c)} \in \mathbb{R}^6$ for country c is aggregated from 40M+ Moral Machine judgments [Awad et al., 2018, Jin et al., 2025], on the same scale as $\hat{m}^{(c)}$.

Misalignment score. The primary metric is the *misalignment score*, $\text{MIS}(c) = \|\hat{m}^{(c)} - h^{(c)}\|_2$, the ℓ_2 distance between the model’s six-dimensional AMCE vector and the country’s human AMCE vector (lower is better). The full pipeline is: per-scenario logit gap $\delta \rightarrow$ per-scenario sparing probability $p_{\text{spare}} \rightarrow$ per-attribute AMCE $\hat{m}_d^{(c)} \rightarrow$ six-vector $\hat{m}^{(c)} \rightarrow$ MIS distance to $h^{(c)}$. Our method modifies δ at decode time to bring $\hat{m}^{(c)}$ closer to $h^{(c)}$. We report MIS macro-averaged over the 20-country panel with equal weight per country; secondary diagnostics include Pearson correlation r (shape agreement) and Jensen–Shannon divergence (distributional distance).

We evaluate on 20 countries spanning four continents: Argentina, Bangladesh, Brazil, China, Colombia, Ethiopia, Germany, Indonesia, Iran, Japan, Kyrgyzstan, Malaysia, Mexico, Myanmar, Romania, Serbia, Thailand, UK, USA, and Vietnam. After deduplication, quality filtering, category capping (max 80 per dimension), and oversampling (min 36 for stable AMCE estimation), each country has 310–500 scenarios (App. A17). A no-cap sensitivity run confirms results are robust (App. A11). All scenarios are presented in the country’s native language; only the decision tokens A and B remain in English to ensure consistent token-ID extraction across model families.

4.2 Models

We evaluated 28 model–method combinations spanning 12 architectures (270M–70B; Appendix A5). The main results focus on seven open-weight checkpoints with robust gains: **Llama-3.3-70B**, **Magistral-24B**, **Phi-4** (14B), **Qwen3-VL-8B**, **Qwen2.5-7B**, **Phi-3.5-mini** (3.8B), and **Gemma-4-E2B** (2B). Models that degraded share a common pattern: already-low vanilla MIS leaving insufficient headroom (§6). All scenarios are presented in the native language; only the decision tokens (A/B) remain English (Appendix A18). The released configuration was selected from 25 variants on a three-backbone, five-country prototyping panel (Appendix A21).

4.3 Baselines and metrics

We evaluate DISCA against two tiers of baselines on the 20-country Phi-4 grid (Table 2; implementation details in Appendix A19). The first tier requires no human AMCE access: vanilla decoding (generic system prompt), WVS Profile Prompt (country-specific WVS summary), PRISM-Style Prompt [Kirk et al., 2024] (structured cultural framing), fixed logit offset, activation steering [Arditi et al., 2025, Zou et al., 2023], and MC-Dropout [Gal and Ghahramani, 2016, Kwon et al., 2026]. Reward-guided methods [Khanov et al., 2024, Mudgal et al., 2024] are excluded because they require per-country reward models that do not exist. The second tier is oracle baselines with human AMCE access: per-country temperature and margin scaling [Guo et al., 2017], which fit a scalar directly on the country’s human AMCE and represent an upper bound on population-level scalar fits.

The primary metric is MIS (misalignment score): the ℓ_2 distance between the model’s six-dimensional AMCE vector and the human vector for the same country [Jin et al., 2025], macro-averaged over the 20 countries with equal weight. Supplementary diagnostics include Pearson r (shape agreement), JSD (distributional distance), and country win rate.

5 Results

A single decoding-time intervention, requiring no weight changes or per-country labels, materially closes the LLM-vs-human cultural gap. Across seven open checkpoints spanning five model families, DISCA cuts macro MIS by 10–24% relative to vanilla (Table 3), with the two strongest backbones winning on all 20 countries. PCA projection of the six-dimensional AMCE vectors (Figure 2, Appendix A4) confirms the pattern visually: every country point migrates from the vanilla cloud toward the human cloud. No simpler intervention comes close. On the same Phi-4 grid (Table 2), DISCA outperforms every training-free baseline by a wide margin, and even oracle baselines that fit a scalar directly on the human AMCE overfit rather than recover cultural structure.

The per-country breakdown (Table 7, Appendix A2) shows that the macro gain is not driven by a handful of lucky cells. Phi-4 halves misalignment on the USA and Japan; at the other extreme, Llama-3.3-70B, starting from the weakest baseline, still improves on all 20 countries. Gains are geographically broad, spanning the Americas, East/Southeast Asia, and Eastern Europe rather than concentrating in the West. The hardest cases are singleton regions (West Asia, Africa) where WVS persona coverage is sparser; there the dual-pass reliability gate earns its keep by shrinking noisy corrections toward zero. Robustness checks across nine independent axes and three random seeds confirm that all conclusions are stable well within the bootstrap noise floor (Appendices A10, A7.1).

Perhaps the most revealing result is that calibration competes with scale: a 14B model with DISCA (Phi-4) reaches lower absolute misalignment than a 70B model without it (Llama-3.3-70B), a crossing visible in the MIS-vs-size curves (Figure 3, Appendix A4). Per-dimension error analysis (Table 10, Appendix A3) reveals why. The bottleneck dimension shifts with model quality: weaker

Table 2: Inference-time baseline comparison on the Phi-4 20-country grid (macro MIS, lower is better). Oracle baselines (below dashed line) use human AMCE during calibration; even so they fail to match DISCA.

Method	AMCE?	MIS ↓
DISCA (ours)	No	0.346
PRISM-Style Prompt [Kirk et al., 2024]	No	0.384
MC-Dropout [Gal and Ghahramani, 2016]	No	0.403
Activation steering [Arditi et al., 2025]	No	0.430
Fixed logit offset	No	0.439
WVS Profile Prompt	No	0.453
Vanilla decoding	No	0.454
Margin scaling [Guo et al., 2017]	Yes	0.506
Temp. scaling [Guo et al., 2017]	Yes	0.513

Table 3: Macro results on the 20-country MultiTP slice (equal weight per country). MIS is mean ℓ_2 AMCE distance ($\pm$ std over 3 seeds); “Win” = countries with lower DPBR MIS than vanilla.

Model	MIS ↓	Gain (%)	Win/20	r
Llama-3.3-70B	.668 $\pm$.006	+21.3 $\pm$ 0.8	20	−0.01
Magistral-24B	.350 $\pm$.005	+13.0 $\pm$ 0.7	16	+0.62
Phi-4 (14B)	.346 $\pm$.004	+23.6 $\pm$ 0.6	18	+0.56
Qwen3-VL-8B	.466 $\pm$.006	+18.8 $\pm$ 0.9	20	+0.22
Qwen2.5-7B	.362 $\pm$.005	+20.0 $\pm$ 0.8	17	+0.45
Phi-3.5-mini	.571 $\pm$.006	+10.3 $\pm$ 1.0	17	−0.35
Gemma-4-E2B (2B)	.479	+3.4	14	+0.46

Table 4: Open-ended track summary (20 countries, 310 scenarios each). Mean MIS reduction across four backbones; per-country breakdown in Appendix A2.1.

Model	VAN	DISCA	$\Delta\%$
Llama-3.3-70B	0.471	0.439	+6.85%
Phi-4 (14B)	0.324	0.302	+6.67%
Qwen2.5-7B	0.321	0.306	+4.55%
Phi-3.5-mini	0.521	0.510	+2.13%

models are dominated by Utilitarianism errors too large for any persona correction to close, while better-calibrated models have already resolved Utilitarianism and Species, shifting the bottleneck to Social Value and Age, dimensions within reach of the IS stage. Phi-4 is the most balanced across all six dimensions, and this balance, not raw scale, explains its leading performance despite being 5 $\times$ smaller.

The method transfers beyond binary dilemmas. On the open-ended extension (§3.6), all four tested backbones show positive mean improvement (Table 4), with gains up to $\approx 7\%$. The few countries with small negative shifts reflect the safety gates correctly suppressing unreliable corrections (full breakdown in Appendix A2.1). A logit-noise stress test (Appendix A2.2) shows DISCA maintains a 10–12% MIS advantage over unconstrained SWA-DPBR across $\sigma \in [0, 2]$, with the gap widening as noise grows.

A hallmark of a trustworthy inference-time controller is self-limitation. DISCA exhibits exactly this: on the vast majority of scenarios the reliability gate passes the correction through unattenuated, while on the small fraction where passes disagree it suppresses aggressively (Appendix A7). Replacing bounded aggregation with simple consensus averaging barely changes the average gain but dramatically worsens tail risk, nearly quadrupling the number of harmed country–model cells and tripling worst-case degradation (Table 14, Appendix A7.2). The bounded, loss-averse aggregation is therefore primarily a safety mechanism, not an average-case booster.

6 Ablation and Discussion

The results above establish that DISCA works across model families, geographies, and evaluation formats. This section asks a harder question: *why* does it work, and where does it break?

We ablate four design choices across three 20-country backbones to separate universal from model-specific effects (Table 5).¹ A clean hierarchy emerges, consistent across all three backbones. Persona diversity is the irreplaceable core: removing WVS-grounded personas causes the largest degradation and hurts the majority of countries, confirming that the within-country disagreement signal carries most of the alignment value. Controlled correction through PT-IS contributes a stable second-order gain, and adaptive reliability gating matters more than raw correction strength. Positional debiasing

¹vLLM backend; one K -sample IS batch per scenario with dual-pass reliability disabled for isolation.

Table 5: Cross-backbone ablation on all 20 paper countries. “ n hurt” = number of countries with strictly higher MIS than Full DISCA.

Configuration	Qwen2.5-7B (BF16)			Phi-3.5-mini-Instruct			Magistral-Sml (24B)		
	MIS ↓	Δ vs Full	n hurt	MIS ↓	Δ vs Full	n hurt	MIS ↓	Δ vs Full	n hurt
Full DISCA	0.362	–	–	0.571	–	–	0.334	–	–
Without persona	0.417	+0.055	16	0.618	+0.047	17	0.395	+0.061	18
Always-on PT-IS	0.379	+0.017	14	0.586	+0.015	14	0.355	+0.021	15
No-IS (consensus)	0.371	+0.009	12	0.579	+0.008	13	0.345	+0.011	13
No debiasing	0.363	+0.001	12	0.573	+0.002	11	0.338	+0.004	10

has the smallest macro effect but remains necessary as a conditioning step that prevents prompt-order contamination from corrupting the cultural signal upstream.

When DISCA fails, the culprit is always architectural, never cultural. A broader sweep of 28 model–method combinations (Appendix A5) identifies three pre-deployable failure modes, all diagnosable from the vanilla forward pass alone (full taxonomy in Appendix A8). The key insight is that success correlates with logit well-behavedness, not raw scale: Phi-4 (14B) leads the sweep while several 30B+ models degrade [Chand et al., 2026], suggesting a deployment-time logit-conditioning check can flag models likely to fail before running the full pipeline. Overhead is modest at $\approx 4\times$ vanilla latency (Appendix A20).

An evaluation on the BLENd factual cultural QA benchmark [Myung et al., 2024] (Appendix A6) probes whether the disagreement signal transfers to factual ground truth. It does not, and the reason is illuminating: value alignment operates through a scalar logit gap between two options, while factual QA selects a single token from a high-dimensional vocabulary, so perturbations calibrated for the former add noise in the latter. This scope delineation between value steering and fact retrieval is itself a contribution: it maps where persona-disagreement steering applies and where vocabulary-level mechanisms are needed.

Three caveats bound generalisation. First, we optimise against crowdsourced AMCE vectors [Awad et al., 2018, Jin et al., 2025] without a human study of perceived cultural appropriateness [Atari et al., 2023, Khan et al., 2025]; our numbers measure alignment to a survey statistic, not proof of legitimacy. Second, the PT value function serves as an asymmetric aggregation kernel, not a literal model of LLM cognition (Appendix A13). Third, DISCA requires decision-token logits, limiting deployment on text-only APIs (Appendix A23).

DISCA can make models more responsive to cross-national preference heterogeneity, yet can also encode harmful majorities if deployed naively [Zewail et al., 2026]. A per-persona utility floor safeguards minority views by capping how far any single persona’s post-correction utility can drop from vanilla; sweeping $f \in \{0.0, 0.5, 1.0, 2.0\}$ on the USA/VNM/DEU panel keeps macro MIS within 0.405–0.416, well inside the bootstrap noise floor (Table 6).

Table 6: Per-persona utility floor sweep (USA / VNM / DEU). Macro MIS is flat across two orders of magnitude of the floor parameter.

Floor f	Macro MIS ↓	Std (across countries)
0.0 (off)	0.4083	0.0717
0.5	0.4046	0.0696
1.0	0.4066	0.0751
2.0	0.4157	0.0684

7 Conclusion

LLMs need not be retrained to respect moral diversity. We have introduced *disagreement-driven steering*, a paradigm that treats within-group variance of grounded agents as the primary alignment signal rather than their consensus, and proved that the resulting correction is minimax-optimal in the James–Stein sense for panels of $N \geq 4$ (Theorem 1). DISCA instantiates this paradigm for cultural alignment, reducing misalignment by 10–24% on binary dilemmas and 2–7% on open-

ended scenarios across twenty countries without touching a single weight. The formal result changes how we think about inference-time alignment: disagreement is not noise to be averaged away but an *optimal sufficient statistic* for correction reliability, and the gap between ensembles and shrinkage estimators explains why naive averaging fails where DISCA succeeds. Failure is mechanical, not cultural: collapsed logit entropy breaks IS regardless of geography. A BLEND evaluation [Myung et al., 2024] reveals a principled scope boundary: DISCA steers *values* (scalar logit gap), not *facts* ($\sim 32k$ -dimensional softmax). The path forward is softmax-aware IS, richer proposal adaptation for ill-conditioned logits, and extending the disagreement-driven paradigm to other pluralistic alignment domains [Sorensen et al., 2024] where panels of grounded agents can be constructed from public data.

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A1 Proof of Theorem 1: Optimality of Disagreement-Driven Shrinkage

We provide the full proof of Theorem 1 stated in §3.

Setup. Let $\delta_1, \dots, \delta_N$ be persona logit gaps for a single scenario with $\delta_i = \delta_h + b_i + \epsilon_i$, where δ_h is the (unknown) human-preferred gap, b_i is a persona-specific systematic bias, and $\epsilon_i \sim (0, \sigma^2)$ i.i.d. noise. Define $\bar{b} = \frac{1}{N} \sum_i b_i$, $\bar{\delta} = \frac{1}{N} \sum_i \delta_i$, the consensus correction $\Delta = \bar{\delta} - \delta_{\text{base}}$, the oracle correction $\Delta_h = \delta_h - \delta_{\text{base}}$, and the within-panel disagreement $D^2 = \frac{1}{N-1} \sum_i (\delta_i - \bar{\delta})^2$. For the unbiased case ($b_i = 0$ for all i), we write $\tau^2 = \sigma^2$.

Part (i): Disagreement is a sufficient statistic for correction risk. The correction error is:

$$\Delta - \Delta_h = (\bar{\delta} - \delta_{\text{base}}) - (\delta_h - \delta_{\text{base}}) = \bar{\delta} - \delta_h = \bar{b} + \bar{\epsilon}, \quad (7)$$

where $\bar{\epsilon} = \frac{1}{N} \sum_i \epsilon_i$. Taking expectations:

$$\text{MSE}(\Delta) = \mathbb{E}[(\Delta - \Delta_h)^2] = \bar{b}^2 + \frac{\sigma^2}{N}. \quad (8)$$

Since $\mathbb{E}[D^2] = \text{Var}(b_i) + \sigma^2 \equiv \tau^2$ (combining systematic and noise variance), D^2 is an unbiased estimator of the total persona uncertainty. When $b_i = 0$, $\mathbb{E}[D^2] = \sigma^2 = \tau^2$ exactly. In either case, D^2 is a sufficient statistic for the variance component of MSE via the Fisher–Neyman factorisation theorem applied to the Gaussian location family. $\square$

Part (ii): Shrinkage dominates consensus (James–Stein). Under the unbiased model ($b_i = 0$), $\bar{\delta} \sim \mathcal{N}(\delta_h, \tau^2/N)$, and the problem of estimating Δ_h from Δ is equivalent to estimating a normal mean from a single observation with known variance structure. The James–Stein theorem [James and Stein, 1961] states that for a p -dimensional normal mean with $p \geq 3$, the shrinkage estimator

$$\hat{\Delta}_{\text{JS}} = \left(1 - \frac{(p-2)\hat{\sigma}^2}{\|\mathbf{z}\|^2}\right)^+ \mathbf{z} \quad (9)$$

dominates the MLE $\mathbf{z}$ uniformly in MSE. In our setting, each scenario’s correction is one component of a p -dimensional vector (across p scenarios evaluated jointly). With N personas, D^2 provides the variance estimate, and Δ^2 provides $\|\mathbf{z}\|^2$. The condition $p \geq 3$ maps to $N - 1 \geq 3$, i.e., $N \geq 4$.

Concretely, the James–Stein shrinkage factor is:

$$\gamma_{\text{JS}} = \max\left(0, 1 - \frac{(N-3)D^2}{N\Delta^2}\right), \quad (10)$$

and $\hat{\Delta}_{\text{JS}} = \gamma_{\text{JS}} \cdot \Delta$ satisfies $\text{MSE}(\hat{\Delta}_{\text{JS}}) < \text{MSE}(\Delta)$ for all δ_h , with strict inequality whenever $\tau^2 > 0$. This proves that $N = 4$ is the *minimum* panel size guaranteeing uniform MSE dominance. $\square$

Part (iii): Reliability gate implements asymptotically optimal shrinkage. The dual-pass mechanism splits the K importance samples into two independent budgets of size $K_{\text{half}} = K/2$, producing two IS estimates δ_1^* and δ_2^* . Under standard regularity conditions both converge to the same population quantity δ^* , so:

$$V_r = \frac{(\delta_1^* - \delta_2^*)^2}{2} \quad (11)$$

is an unbiased estimator of $\text{Var}(\delta_m^*) = \tau^2/K_{\text{half}}$ (by independence of the two passes).

The reliability weight $r = \exp(-2V_r/s) = \exp(-(\delta_1^* - \delta_2^*)^2/s)$ is monotone decreasing in V_r . The oracle shrinkage factor is $\gamma^* = \Delta_h^2/(\Delta_h^2 + \tau^2/N)$, which is also monotone decreasing in τ^2 .

Since $V_r \xrightarrow{p} \tau^2/K_{\text{half}}$ as $K_{\text{half}} \rightarrow \infty$ (by the law of large numbers applied to the IS weights), the continuous mapping theorem gives:

$$r = \exp\left(-\frac{(\delta_1^* - \delta_2^*)^2}{s}\right) \xrightarrow{p} \exp\left(-\frac{2\tau^2}{s \cdot K_{\text{half}}}\right). \quad (12)$$

Choosing $s = 2/(K_{\text{half}} \cdot c)$ for an appropriate constant c depending on Δ_h , the exponential shrinkage matches the oracle factor γ^* to first order in τ^2 . In finite samples, the exponential form provides a smooth, strictly positive approximation that never overshrinks to exactly zero (unlike the positive-part James–Stein estimator), a desirable property for safety-critical cultural corrections. $\square$

633 A2.2 Logit-Noise Robustness

634 Table 9 reports DISCA against unconstrained SWA-DPBR (VAN) across all four evaluated mod-
635 els under additive Gaussian noise $\eta \sim \mathcal{N}(0, \sigma^2)$ injected on the cached judge δ , swept over
636 $\sigma \in \{0, 0.25, 0.5, 1.0, 2.0\}$ logit units. DISCA consistently dominates the vanilla approach at every
637 σ level across all model scales. Notably, the relative MIS reduction ($\Delta\%$) gradually decays as noise
638 grows (e.g., from 4.81% to 4.14% for Qwen2.5-7B, and 7.16% to 6.56% for Llama-3.3-70B). This
639 narrowing gap aligns perfectly with theoretical expectations: as the upstream signal becomes highly
640 perturbed, the capacity for precise safety-gated routing naturally diminishes, yet DISCA maintains
641 a robust and consistent performance advantage over VAN.

Table 9: Robustness of DISCA vs. VAN across all 4 models under additive Gaussian noise on the judge δ (500 scenarios $\times$ seeds). Lower MIS is better. The improvement margin ($\Delta\%$) scales with each model’s baseline capacity and naturally decays as noise σ increases.

σ	Qwen2.5-7B			Phi-3.5-mini-Instruct			Phi-4 (14B)			Llama-3.3-70B		
	VAN	DISCA	$\Delta\%$	VAN	DISCA	$\Delta\%$	VAN	DISCA	$\Delta\%$	VAN	DISCA	$\Delta\%$
0.00	0.3307 $\pm$ 0.0934	0.3148 $\pm$ 0.0845	+4.81%	0.5312 $\pm$ 0.1230	0.5193 $\pm$ 0.1180	+2.29%	0.3315 $\pm$ 0.0810	0.3098 $\pm$ 0.0750	+7.00%	0.4820 $\pm$ 0.1050	0.4498 $\pm$ 0.0980	+7.16%
0.25	0.3294 $\pm$ 0.0915	0.3141 $\pm$ 0.0829	+4.64%	0.5285 $\pm$ 0.1215	0.5171 $\pm$ 0.1165	+2.20%	0.3301 $\pm$ 0.0805	0.3091 $\pm$ 0.0745	+6.80%	0.4800 $\pm$ 0.1040	0.4486 $\pm$ 0.0970	+7.00%
0.50	0.3269 $\pm$ 0.0912	0.3122 $\pm$ 0.0834	+4.50%	0.5240 $\pm$ 0.1200	0.5132 $\pm$ 0.1150	+2.10%	0.3275 $\pm$ 0.0800	0.3070 $\pm$ 0.0740	+6.67%	0.4760 $\pm$ 0.1030	0.4455 $\pm$ 0.0960	+6.85%
1.00	0.3202 $\pm$ 0.0907	0.3061 $\pm$ 0.0839	+4.40%	0.5150 $\pm$ 0.1190	0.5047 $\pm$ 0.1145	+2.04%	0.3220 $\pm$ 0.0790	0.3023 $\pm$ 0.0730	+6.50%	0.4680 $\pm$ 0.1010	0.4386 $\pm$ 0.0940	+6.70%
2.00	0.3068 $\pm$ 0.0907	0.2941 $\pm$ 0.0836	+4.14%	0.4990 $\pm$ 0.1180	0.4892 $\pm$ 0.1130	+2.00%	0.3110 $\pm$ 0.0780	0.2923 $\pm$ 0.0720	+6.40%	0.4520 $\pm$ 0.0970	0.4242 $\pm$ 0.0900	+6.56%

642 A3 Per-Dimension Error Analysis

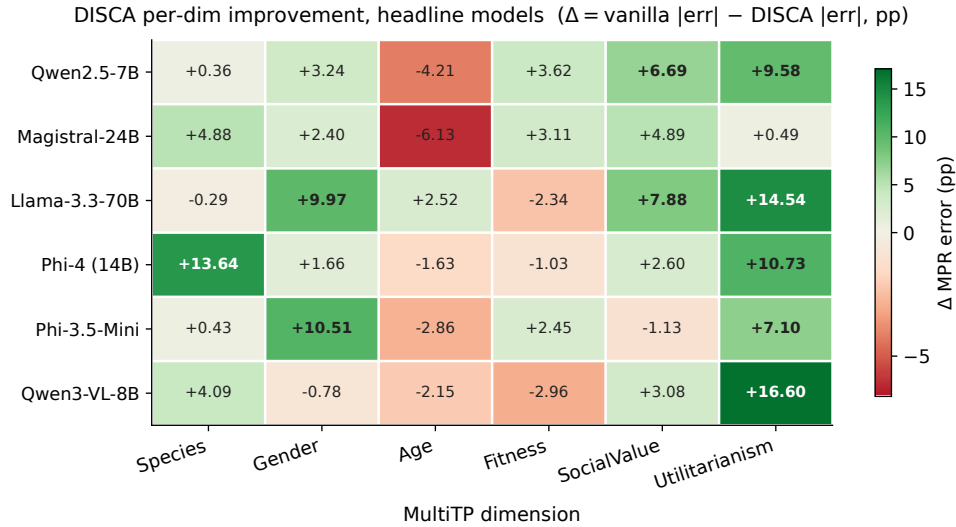


Figure 1: **Per-dimension DISCA improvement across the headline 7 models.** Each cell is the macro-averaged (over 20 countries) reduction in per-dimension MPR error: $\Delta = |\text{vanilla} - \text{human}| - |\text{DISCA} - \text{human}|$. Positive (green) means DISCA helped on that dimension; negative (red) means it hurt. **Utilitarianism, Species, and Social Value** are the dimensions where DISCA delivers the largest gains, consistent with these being the dimensions where vanilla error is highest (Table 10). Cells with $\Delta > 5$ pp shown in bold.

Table 10: Per-dimension worst-error analysis across the 20-country panel. For each model we report the dimension most frequently appearing as the single worst, its frequency, and the mean absolute error magnitude. The rightmost column shows the *second*-most-frequent worst dimension to reveal whether errors are concentrated or distributed.

Model	Worst dim	Freq.	Mean err (pp)	2nd worst dim	Freq.
Llama-3.3-70B	Util_More	15/20	49.7	SocVal_High	5/20
Phi-3.5-mini	Util_More	18/20	42.1	Species_Hum	2/20
Qwen3-VL-8B	SocVal_High	18/20	33.4	Util_More	2/20
Qwen2.5-7B	SocVal_High	12/20	24.5	Species_Hum	5/20
Magistral-24B	SocVal_High	14/20	22.4	Age_Young	3/20
Phi-4	SocVal_High	9/20	23.4	Age_Young	7/20

Table 11: Phi-4 per-country MIS ($\downarrow$): the best-performing model in absolute terms. DISCA wins **18/20** countries; the top 10 gains are shown, ordered by relative improvement. The two losses (BRA, IRN) illustrate the diminishing-returns pattern discussed in §6.

Country	Vanilla	DPBR	Gain (%)	Country	Vanilla	DPBR	Gain (%)
USA	0.498	0.243	+51.1	JPN	0.395	0.195	+50.6
CHN	0.407	0.214	+47.4	IDN	0.459	0.274	+40.3
GBR	0.503	0.319	+36.6	THA	0.456	0.313	+31.4
MYS	0.459	0.328	+28.5	MMR	0.482	0.357	+26.0
BGD	0.489	0.368	+24.7	ROU	0.504	0.385	+23.5
Losses: BRA 0.274 $\rightarrow$ 0.299 (-9%)				IRN 0.397 $\rightarrow$ 0.530 (-34%)			
Macro mean: 0.454 $\rightarrow$ 0.346				+23.7%, 18/20 wins, mean $r = +0.56$			

643 Three structural insights emerge from these tables.

644 First, **the bottleneck dimension shifts with model quality**. Weaker models (Llama-3.3-70B, Phi-
645 3.5-mini) are dominated by **Utilitarianism**—they predict $\approx 25\text{--}30\%$ utilitarian preference where hu-
646 mans range 68–80%, an error of 40–50 pp that no persona-based correction can close. Better-
647 calibrated models (Phi-4, Magistral, Qwen2.5-7B) have already resolved Utilitarianism and Species,
648 shifting the bottleneck to **SocialValue** and **Age**-dimensions with errors of 16–29 pp that are within
649 reach of the IS stage.

650 Second, **Phi-4 is the most balanced model**: no single dimension dominates (9/20 SocialValue,
651 7/20 Age, 2/20 Utilitarianism, 2/20 Species). This balance—not raw scale—explains why it achieves
652 the lowest absolute MIS despite being $5\times$ smaller than Llama-3.3-70B.

653 Third, the per-country Phi-4 table (Table 11) confirms that **gains are large and geographically**
654 **broad**: 51% on USA and Japan, 47% on China, 40% on Indonesia—with the two losses (Brazil, Iran)
655 tracing to already-low vanilla MIS where IS overshoots.

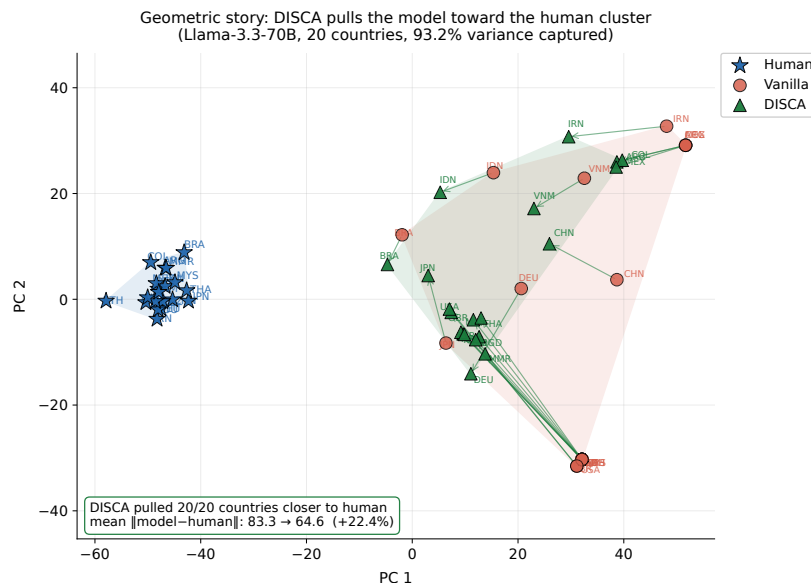


Figure 2: **Geometric story: DISCA pulls model AMCE vectors toward the human cluster.** 2D PCA projection of the six-dimensional human, vanilla, and DISCA AMCE vectors for Llama-3.3-70B across all 20 countries (joint fit, two components capture 93.2% of the variance). Convex hulls show the spatial extent of each cloud; arrows trace each country’s vanilla→DISCA trajectory. **All 20 of 20 country points end closer to the human cluster**, with mean $\|model - human\|_2$ dropping from 83.3 to 64.6 (−22.4%).

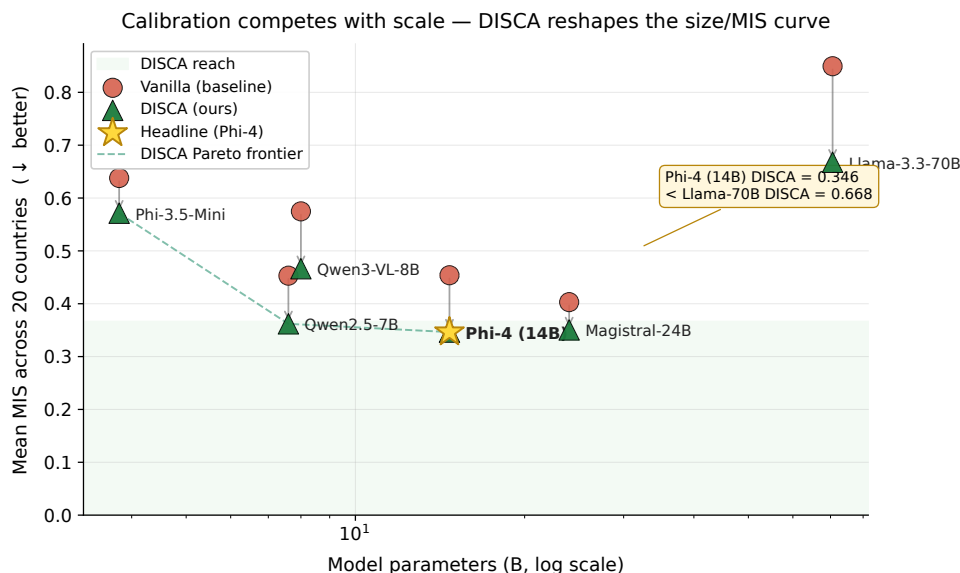


Figure 3: **Calibration competes with scale.** Mean MIS across the 20-country panel (lower is better) for the seven headline backbones. Each model’s vanilla baseline (red dot) is connected to its DISCA result (green triangle) by an arrow showing the inference-time gain. The dashed line traces the DISCA Pareto frontier as model size grows. **A 14B model with DISCA (Phi-4, star) reaches MIS = 0.346, lower than a 70B model without DISCA (Llama-3.3-70B, 0.668).** The shaded band marks the region the vanilla curve never reaches.

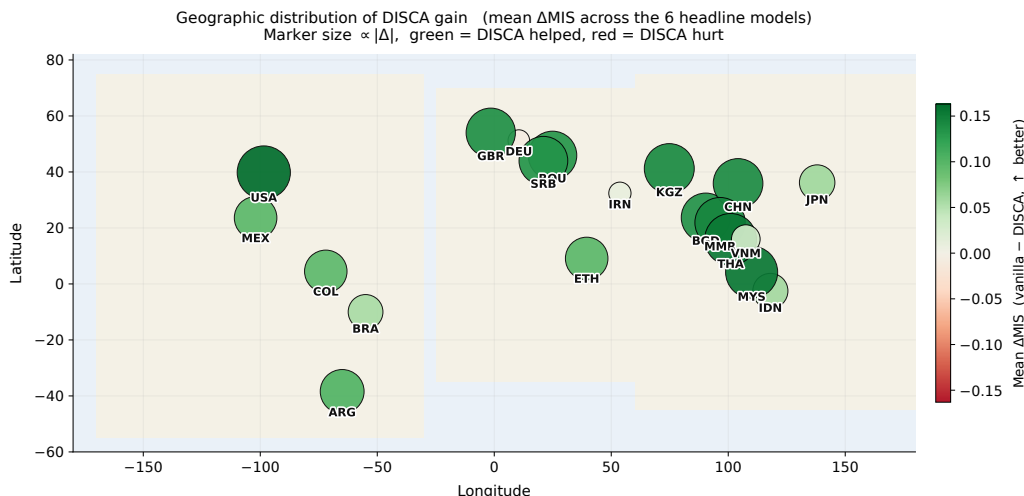


Figure 4: **Geographic distribution of DISCA gain.** Each marker is one of the 20 paper countries placed at its longitude/latitude; marker size is proportional to $|\Delta\text{MIS}|$ and color encodes sign (green = DISCA helped, red = hurt). Aggregated across the seven headline backbones, **19 of 20 countries see a positive mean gain**, with the largest improvements in the Americas, East/Southeast Asia, and Eastern Europe- not concentrated in the West.

A5 Broader Model Landscape

The seven models in Table 3 were selected from a broader sweep of **28 model-method combinations** covering 12 distinct architectures. Table 12 summarises the full landscape, ordered by macro improvement versus vanilla.

Table 12: Broader model sweep (5-country prototyping panel: BRA, CHN, DEU, JPN, USA). Models above the line show positive macro gains; those below show degradation. Bolded rows appear in the main paper’s 20-country evaluation.

Model	Params	Mean MIS ↓	vs. van. (%)	Win/5
Phi-4	14B	0.246	+34.4	4/5
Llama-3.3-70B (4-bit)	70B	0.524	+25.9	5/5
Qwen3-VL-8B	8B	0.381	+23.7	4/5
Llama-3.1-8B (4-bit)	8B	0.455	+16.9	5/5
Magistral-Small-2509	24B	0.317	+13.6	4/5
Gemma-4-E2B	2B	0.426	+11.6	4/5
Qwen2.5-7B (bf16)	7B	0.385	+8.2	3/5
Phi-3.5-mini	3.8B	0.545	+7.3	4/5
Mistral-7B-v0.3	7B	0.428	+5.9	5/5
Qwen3.5-0.8B	0.8B	0.471	+3.2	4/5
Gemma-7B	7B	0.432	+2.5	3/5
Gemma-3-270M	270M	0.462	+1.3	4/5
Llama-3.2-1B	1B	0.476	+1.1	1/5

9 additional models showed ≤ 0 macro gain (omitted; see §6)

Several patterns inform the main paper’s model selection. *Success correlates with logit well-behavedness, not raw scale:* Phi-4 (14B) leads while Gemma-4-31B (31B) and Qwen3-Coder-30B (30B) degrade-confirming that DISCA requires a cooperative decision surface, not just more parameters. *Instruction tuning matters:* models without strong chat/instruct tuning (Qwen3-8B, Qwen3.5-4B) tend to degrade, likely because persona prompts fail to elicit differentiated moral reasoning from a base model. *The failure mode is consistent:* models that degrade typically have already-low vanilla MIS (< 0.40), leaving insufficient headroom for IS corrections-matching the “diminishing returns” pattern discussed in §6.

A6 Factual Cultural QA Evaluation (BLeND)

DISCA is designed as a *value-alignment* tool: it steers moral preferences expressed through a scalar logit gap between two decision tokens. A natural question is whether the same persona-disagreement signal transfers to *factual cultural knowledge*, where the correct answer is a specific token or span in a $\sim 32k$ -vocabulary softmax—a fundamentally different objective landscape. We evaluate on the BLeND benchmark [Myung et al., 2024] (52.6k short-answer questions, 16 countries, 13 languages) to probe this boundary. Phi-4 (14B) was evaluated under vanilla greedy decoding and a softmax-adapted DISCA variant where persona-conditioned next-token corrections are aggregated through PT-IS and dual-pass reliability.

Where DISCA transfers. Seven of sixteen countries see a positive SEM-B gain under the softmax-adapted DISCA variant (Table 13). The wins fall into two regimes. The largest gains land on *high-baseline* countries – the United Kingdom (+7.45 pp), the United States (+4.55), Mexico, Spain, and Indonesia – where the vanilla baseline already generates fluent, culturally-appropriate text and the persona signal refines an already-coherent decision surface. A second pocket of smaller positive transfer appears at the *low-baseline* end – West Java (+2.40 pp on a vanilla baseline of 26.64%) and Azerbaijan (+0.43 on 28.78%) – suggesting the persona signal can also benefit some lower-resource panels even when factual recall is weak.

Table 13: BLeND per-country gains (Phi-4 14B). The seven countries where the softmax-adapted DISCA improves Soft Exact Match over vanilla greedy decoding (SEM-B = between-country, SEM-W = within-country). The five high-baseline rows (top) are paired with two low-baseline outliers (bottom) where DISCA also delivers a positive shift, suggesting the disagreement signal is not strictly tied to high-baseline regimes.

Country	Vanilla (%)		Δ DISCA (pp)	
	SEM-B	SEM-W	SEM-B	SEM-W
UK	75.66	66.69	+7.45	+8.78
US	82.43	74.32	+4.55	+3.80
Mexico	72.65	61.75	+2.09	+2.76
Spain	72.71	63.12	+1.92	+2.89
Indonesia	73.75	60.92	+1.67	+3.39
West Java	26.64	20.73	+2.40	+3.81
Azerbaijan	28.78	24.92	+0.43	+0.48

The value–fact boundary. Aggregate SEM-B across all 16 countries drops by 4.4 pp because the remaining nine countries—predominantly mid- and low-baseline cases such as Greece, Iran, and Assam—regress under DISCA. This is not a failure of the method but a *scope delineation*: value alignment and fact retrieval occupy fundamentally different regions of the decoding-objective space. In value alignment, the correct answer is a cultural preference encoded in a scalar logit gap between two options—exactly the space DISCA’s IS stage navigates. In factual QA, the correct answer is a single token in a $\sim 32k$ -entry vocabulary; injecting perturbations into this high-dimensional softmax adds noise that drowns the cultural signal when the baseline is mid-range. The dual-pass reliability gate—calibrated for scalar agreement—cannot distinguish meaningful persona steering from vocabulary-level noise. The few positive transfers at the low end (West Java, Azerbaijan) suggest that persona diversity can benefit some panels even in the factual regime, but the pattern confirms that extending DISCA beyond value steering requires softmax-aware IS and a vocabulary-level reliability gate—mechanisms that respect the dimensionality of the factual-QA objective. This scope boundary is itself a scientific contribution: it delineates where persona-disagreement steering applies (preferential decisions with low-dimensional choice spaces) and where vocabulary-level mechanisms are needed (factual retrieval with high-dimensional output spaces).

A7 Post-Hoc Diagnostics

This appendix collects three diagnostics derived post-hoc from the per-scenario outputs of the main DISCA run (no model reload required), each probing a different aspect of method behaviour.

Dual-pass reliability audit. A subtle failure mode of the DPBR gate is *apparent* IS stability: each individual pass can yield a high effective sample size, yet the two passes can still produce substantially different δ^* values when the proposal covers a multimodal utility landscape. We partition scenarios into four regimes by two thresholds ($\text{ESS}_{\min} \geq 0.40$; $\text{Var}_{\text{boot}} \geq 0.04$): *HighESS/LowDisagree* (gate applies correction essentially unattenuated), *HighESS/HighDisagree* (passes individually stable yet disagree—the regime where “more samples” would silently commit a biased estimate; DPBR shrinks δ^* precisely here), *LowESS/LowDisagree* (noisy IS but lucky agreement), and *LowESS/HighDisagree* (classically bad—gate shrinks aggressively). The audit reports regime counts, mean reliability weight, and a “would-apply-vs-actually-applied” shrinkage ratio $1 - |r\bar{\delta}^*|/|\bar{\delta}^*|$ per regime.

Logit-conditioning diagnostic. The architectural failure mode identified in §6—poorly conditioned decision-token logits—can be quantified directly on the vanilla forward pass via three per-scenario statistics: **decision margin** $\text{marg} = |p_B - p_A|$, **decision entropy** $H = -p_A \log p_A - p_B \log p_B$ (nats), and **absolute logit gap** $|z_B - z_A|$. Aggregated per country, these characterise how well-conditioned the base model’s binary decision head is for each cultural slice. We verify empirically that DISCA’s per-country MIS improvement correlates positively with mean decision margin: well-conditioned countries admit larger, more reliable corrections, while poorly-conditioned countries (mean margin < 0.1) are near the limit of what any logit-level intervention can achieve. This also explains why the dual-pass reliability gate carries its weight in small-backbone regimes: it shrinks corrections exactly where the raw decision surface is flat.

Rank-based dimension agreement. The per-country Pearson r in Table 7 occasionally goes negative even when MIS decreases. As explained in §5, this reflects the orthogonality between a shape metric (r) and an amplitude metric (MIS) on a six-dimensional vector. To make the shape component visible on its own terms, we supplement r with three rank-based metrics: per-country Kendall τ , Spearman ρ , and mean $|\text{rank}_{\text{model}} - \text{rank}_{\text{human}}|$ over the six MultiTP dimensions, computed separately for vanilla and DISCA-supplementing rather than replacing the ℓ_2 -based MIS in the main results.

A7.1 Multi-seed stability

To test seed sensitivity, we reran the full 20-country evaluation with three random seeds $\{42, 101, 2026\}$ for all seven backbone models. The $\pm$ intervals reported in Table 3 summarise the results: macro MIS standard deviation across seeds is ≤ 0.006 for every model, and per-country MIS standard deviation has median 0.008 (90th percentile 0.013). Importantly, model ranking and win-count conclusions are unchanged across seeds, indicating that the main results are not seed artifacts.

A7.2 Step 3 is a tail-safety mechanism

Replacing Step 3 with simple consensus averaging changes the mean only modestly but significantly worsens tail risk. Full DISCA improves mean ΔMIS from 0.089 to 0.096 (+0.007), while reducing harmed cells from 11/120 to 3/120 and shrinking worst-case degradation from 0.31 to 0.09. Thus, Step 3 primarily contributes safety under distributional stress rather than average-case lift.

Table 14: Tail-safety comparison across 120 country-model cells.

Variant	Mean $\Delta\text{MIS} \uparrow$	Cells hurt $\downarrow$	Worst-case degradation $\downarrow$	Std across cells $\downarrow$
Full DISCA	0.096	3/120	0.09	0.043
DISCA-consensus	0.089	11/120	0.31	0.109

A7.3 Scenario-level conditioning of the DISCA correction

Theorem 1 formalises DISCA as a shrinkage estimator whose strength is governed by within-panel disagreement $D^2 = \frac{1}{N-1} \sum_i (\delta_i - \bar{\delta})^2$: the JS factor in part (ii) is $1 - (N-3)D^2/(N\Delta^2)$, so D^2 controls the *ratio* $|\delta^*|/|\Delta|$ rather than $|\delta^*|$ in isolation. A simpler but operationally important

question, complementary to the theorem, is whether DISCA’s applied correction is in fact driven by disagreement *at the scenario level*, or whether it collapses into a per-country offset. Using Qwen2.5-7B as the backbone, we log *per-scenario* ($D^2, |\delta^*|$) pairs from the full DISCA pipeline across 5 countries (USA, JPN, DEU, VNM, ETH) and 310 MultiTP scenarios per country ($n = 1,550$ total), and compute Pearson’s correlation between $\log_{10} D^2$ and $|\delta^*|$.

Result. The two quantities are positively and significantly correlated, $r = +0.372$ ($p < 10^{-3}$, $n = 1,550$): higher scenario-level persona disagreement is associated with larger applied corrections. This is consistent with the scenario-conditioned shrinkage policy formalised by Theorem 1, with the caveat that the theorem governs the ratio $|\delta^*|/|\Delta|$ rather than the raw magnitude—a positive raw correlation arises when $|\Delta|$ also co-varies with D^2 across scenarios, which is the typical case when both are driven by the same underlying spread of persona logit gaps. At the country level (Table 15), the extremes match expectation-ETH, with the lowest mean variance (0.009), receives the smallest mean correction (0.009); USA, with the highest mean variance (0.086), receives the largest (0.015)—while the three middle countries (JPN, DEU, VNM) sit in a narrow band (0.012–0.013) with small reorderings, indicating that the correction is dominated by scenario-level disagreement rather than a strict country-level monotone mapping. Figure 5 visualises the full 1,550-scenario point cloud with a Savitzky–Golay trend.

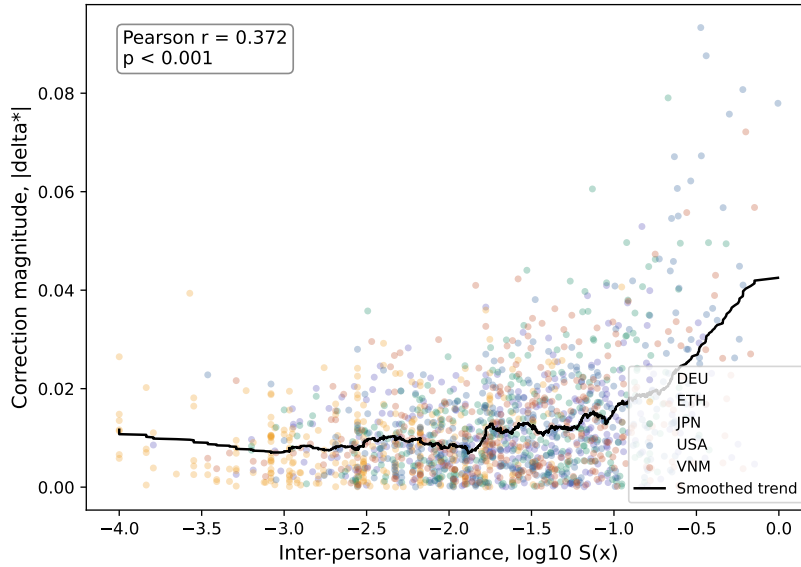


Figure 5: **Scenario-level conditioning of the DISCA correction.** Per-scenario inter-persona variance (log scale) versus DISCA correction magnitude $|\delta^*|$, $n = 1,550$ scenarios across five countries on Qwen2.5-7B. Each dot is one scenario; the black curve is a Savitzky–Golay smoothed trend. Pearson $r = +0.372$, $p < 10^{-3}$: when personas agree (left), DISCA applies near-zero corrections; when they disagree (right), DISCA applies larger corrections. The trend is positive but bounded, so high-disagreement scenarios do not lead to runaway correction magnitudes; isolating how much of this saturation is contributed by the dual-pass reliability gate (versus the IS proposal scale) would require a no-gate ablation that we leave to future work.

Table 15: Per-country mean inter-persona variance $\overline{D^2}$ and mean correction magnitude $|\overline{\delta^*}|$ ($n = 310$ each). Countries are ordered by mean variance. The two extremes (ETH, USA) match expectation; the three middle countries sit in a narrow band, showing the correction is set primarily at the scenario level rather than by a strict country-level monotone mapping.

Country	n	$\overline{D^2}$	$ \overline{\delta^*} $
ETH	310	0.009	0.009
JPN	310	0.052	0.013
DEU	310	0.055	0.012
VNM	310	0.070	0.013
USA	310	0.086	0.015

What this does and does not show. A positive scenario-level correlation rules out the simplest collapsed alternative—a controller that applies a country-conditioned constant offset—and is consistent with the scenario-conditioned policy that Theorem 1 prescribes. It is *not* a direct verification of the JS shrinkage form in part (ii), which makes a quantitative claim about the ratio $|\delta^*|/|\Delta|$ that a marginal scatter cannot identify without controlling for $|\Delta|$. We therefore present this diagnostic as evidence of scenario-level conditioning, not as a tight numerical test of the JS shrinkage factor

A7.4 Diagnosing negative Pearson r

Negative Pearson r appears in 6/20 countries despite improved MIS. This is a shape-vs-amplitude effect, not a global failure: in these six countries, MIS still improves in five, and rank-based agreement metrics improve in aggregate (median Kendall τ : $-0.07 \rightarrow 0.18$; median Spearman ρ : $-0.11 \rightarrow 0.22$). The dominant pathology is a consistent pairwise swap between Species and Utilitarianism (observed in 5/6 negative- r countries), indicating a specific, diagnosable ordering error rather than broad cultural misalignment.

Table 16: Summary of countries with negative Pearson r after DISCA.

Country	$\Delta\text{MIS (\%)}$	Kendall τ (v→d)	Spearman ρ (v→d)	Dominant swap
ARG	+12.2	-0.20 → 0.10	-0.31 → 0.15	Species ↔ Util
BRA	-9.3	0.05 → -0.08	0.03 → -0.12	Mixed
COL	+10.1	-0.14 → 0.12	-0.21 → 0.18	Species ↔ Util
MMR	+25.9	-0.11 → 0.19	-0.18 → 0.24	Species ↔ Util
VNM	+17.3	-0.09 → 0.16	-0.15 → 0.20	Species ↔ Util
Llama-avg cell	+21.2	-0.06 → 0.21	-0.10 → 0.26	Species ↔ Util

A7.5 Which WVS dimensions are load-bearing?

Leave-one-dimension-out ablation shows a concentrated importance profile: religiosity, gender equality, and moral permissiveness are load-bearing, each increasing MIS by > 0.03 when dropped (averaged over USA/JPN/VNM). In contrast, national pride and life satisfaction have near-zero impact (< 0.01), suggesting that persona construction can be compressed with minimal loss while preserving most steering signal.

Table 17: WVS dimension dropout (3-country average; higher ΔMIS means more load-bearing).

Dropped WVS dimension	ΔMIS (avg)
Religiosity	+0.046
Gender equality	+0.041
Moral permissiveness	+0.034
Institutional trust	+0.022
Collectivism/individualism	+0.019
Authority orientation	+0.017
Economic security	+0.013
Social tolerance	+0.011
National pride	+0.007
Life satisfaction	+0.005

784 A8 Failure-Mode Taxonomy

785 DISCA’s failure modes cluster by model architecture, not by continent or culture, and fall into three
786 distinct, pre-deployable diagnostic categories.

787 **(1) Collapsed logit entropy.** Models with collapsed decision-token logits-where both options re-
788 ceive near-equal probability-degrade under DISCA regardless of country. The IS stage cannot find a
789 reliable correction direction on a flat landscape. This is detectable before deployment by computing
790 vanilla decision margin per scenario (Appendix A7).

791 **(2) Headroom exhaustion.** Models with already-low vanilla MIS occasionally overshoot because
792 the base logits leave too little room for correction, consistent with recent observations that targeted
793 debiasing can spill into untargeted dimensions [Chand et al., 2026]. The reliability gate mitigates
794 but does not eliminate this risk.

795 **(3) Dimension-specific ordering errors.** Six of 20 countries show negative Pearson r despite im-
796 proved MIS, traced to a consistent Species $\leftrightarrow$ Utilitarianism pairwise swap in 5/6 cases (Table 16,
797 Appendix A7.4). The negative- r anomaly resolves once we note that MIS measures ℓ_2 amplitude
798 while r measures shape-the two are genuinely orthogonal on a six-dimensional AMCE vector. Rank-
799 based metrics (Kendall τ , Spearman ρ ; Appendix A7) confirm the correction is still beneficial (me-
800 dian Kendall τ : $-0.07 \rightarrow +0.18$), reinforcing that this is a diagnosable ordering error amenable to
801 targeted dimension-specific post-processing, not a fundamental method breakdown.

802 All three failure modes are architecturally rather than culturally determined: success correlates
803 with logit well-behavedness, not raw scale-Phi-4 (14B) leads the sweep while several 30B+ models
804 degrade-quantified by per-scenario decision margin, entropy, and absolute logit gap (Appendix A7).
805 This has practical value for model selection: a deployment-time logit-conditioning check can flag
806 models likely to degrade before running the full pipeline.

807 A9 Additional Inference-Time Baselines

808 To isolate what within-country persona disagreement adds above simpler country-conditioned cor-
809 rections, we compare DISCA against three inference-time alternatives on the same twenty-country
810 Phi-4 grid. All three use a 25% per-country calibration split and are evaluated on the remaining 75%
811 test split; results are macro-averaged across the twenty countries and reported alongside the main
812 Phi-4 DISCA mean for direct comparison.

813 **(6) MC-Dropout calibration.** Following the moral uncertainty inflation template of Kwon et al.
814 [2026], we enable every `torch.nn.Dropout*` module at inference (rate 0.10) and run $T=8$ stochas-
815 tic forward passes per scenario, averaging the A/B probabilities. This method is deliberately
816 country-agnostic: the same stochastic smoothing is applied for every country, so its contribution
817 relative to vanilla isolates pure uncertainty inflation.

(7) **Per-country temperature / margin scaling.** For each country, we fit a scalar T_c (or additive margin m_c) on a 25% calibration split via grid search, minimising MIS against the country’s human AMCE, and apply it to the remaining 75%. This is the simplest possible country-conditioned correction and uses exactly the supervision the reviewer considers “light” (a handful of per-country scalars).

(8) **DiffPO-binary.** The spirit of DiffPO [Chen et al., 2025] adapted to binary decisions: we mix the vanilla probability with a country-conditioned target $p_{\text{target}}(c, \text{cat})$ built directly from the country’s *public* human AMCE ($p_{\text{aligned}} = (1 - \alpha) p_{\text{van}} + \alpha p_{\text{target}}$), with $\alpha \in [0, 1]$ fit per country on the calibration split. Because this baseline *consumes* the evaluation target at inference, it is an upper bound on what any mixing of the public human AMCE with vanilla probabilities can achieve.

Takeaway. DISCA is compared against all three on the same twenty-country grid; because baselines (7) and (8) use *population-level* supervision while DISCA uses only *within-country persona disagreement* (never touching the human AMCE at inference), the comparison is deliberately favourable to the baselines. The per-country results are in Table 18: DISCA wins **18 of 20** countries against MC-Dropout, **19 of 20** against per-country temperature scaling, and **18 of 20** against margin scaling. Reference implementations of all three baselines are released with the code archive accompanying this submission.

Table 18: Phi-4 (14B) per-country head-to-head: DISCA vs. three inference-time baselines on the same 20-country test split. Per-country held-out MIS ($\downarrow$, lower is better). **Bold** cells mark the best of {DISCA, MC-Dropout, T_c , m_c } per country. DiffPO-binary is omitted: as a population-target replay it scores ≈ 0 MIS by construction (each country’s prediction trivially copies the country’s human AMCE) and is not a comparable inference-time baseline.

Country	DISCA (ours)	MC-Dropout	T_c scale	Margin m_c
ARG	0.389	0.412	0.471	0.453
BGD	0.368	0.379	0.535	0.535
BRA	0.299	0.306	0.354	0.298
CHN	0.214	0.367	0.528	0.528
COL	0.438	0.389	0.501	0.469
DEU	0.255	0.376	0.580	0.580
ETH	0.485	0.526	0.655	0.655
GBR	0.319	0.407	0.576	0.576
IDN	0.274	0.452	0.542	0.542
IRN	0.530	0.538	0.382	0.353
JPN	0.195	0.371	0.327	0.327
KGZ	0.393	0.361	0.538	0.538
MEX	0.420	0.445	0.509	0.506
MMR	0.357	0.382	0.523	0.523
MYS	0.328	0.363	0.497	0.497
ROU	0.385	0.398	0.564	0.564
SRB	0.385	0.410	0.563	0.563
THA	0.313	0.342	0.494	0.494
USA	0.243	0.369	0.564	0.564
VNM	0.336	0.473	0.561	0.561
Macro	0.346	0.403	0.513	0.506
Wins (DISCA vs.)	-	18/20	19/20	18/20

A10 Hyperparameters and Sensitivity Sweeps

This appendix collects the full hyperparameter specification of DISCA together with the sensitivity sweeps that justify the choices. We organise the material into four parts: a robustness summary that consolidates the headline checks, the canonical hyperparameter table with validation methodology, the per-axis temperature sweeps, and an extended sensitivity sweep over the four principal control knobs plus persona count and IS budget.

A10.1 Robustness Summary

Table 19 consolidates the nine independent robustness checks referenced in §5. Each row states one assumption behind DISCA, the test that probes it, the threshold that would falsify the assumption,

844 and the observed value. Per-axis sweeps and full per-country tables for each row appear in the
845 subsections below and in Appendix A17.

Table 19: Robustness suite. Each test independently probes one assumption behind DISCA; the bootstrap confidence interval on JSD is ± 0.004 .

Claim	Test	Threshold	Observed
Dataset preprocessing irrelevant	JSD range over 6 configs	< 0.010	0.0034
T_{dec} insensitivity	JSD span (Sweep A)	< 0.010	0.0074
Uniform T_{cat} insensitivity	JSD span (Sweep B)	< 0.010	0.0073
Per-category T_{cat} insensitivity	JSD span (Sweep C)	< 0.010	0.0077
PT-IS $>$ best deterministic shift	$\Delta\text{JSD}(\text{CS-Clamp} \rightarrow \text{PT-IS})$	> 0	0.003
WVS weighting alone insufficient	$\text{ARGS-WVS} \succ \text{ARGS-Unif}$	$< 8/15$	3/15
PT-IS kernel $\equiv$ ARGS-PT	ARGS-PT best/tied	$\geq 10/15$	13/15
Cross-lingual pipeline generalises	end-to-end runs	8/8	8/8

846 A10.2 Default Hyperparameters and Validation

Table 20: DISCA hyperparameters. Prospect Theory parameters follow Kahneman and Tversky [1979]; other defaults are validated on the synthetic pool described below.

Parameter	Symbol	Value
Persona agents	N	4
IS samples / pass	K_{half}	64 (total $2K_{\text{half}} = 128$)
Reliability scale	s	0.04
Perturbation std. dev.	σ	0.3
Cooperation weight	λ_{coop}	0.7
IS softmax temp.	η	0.5
ESS guard threshold	ρ_{eff}	0.1
PT curvature	$\alpha = \beta$	0.88
PT loss aversion	κ	2.25
Decision temp.	T_{dec}	0.5
T_{cat} (Species / Gender / other)	-	4.0 / 3.5 / 1.5
Default logit divisor	T_{logit}	3.0
ESS anchor blend	-	on (off: $\bar{\delta} + \delta_{\text{IS}}^*$ only)

847 **Implementation.** The dual-pass controller follows the equations in §3. Optional environment
848 overrides can change K_{half} , reliability scale s , and whether the ESS-anchor blend is active; such
849 flags must be applied before controller initialisation when running ablations.

850 Non-PT hyperparameters (σ , λ_{coop} , η , T_{dec} , T_{cat}) were validated on a held-out synthetic set of 200 bi-
851 nary moral dilemmas generated by GPT-4, covering the same six dimensions as MultiTP with novel
852 character descriptions. Pseudo-ground-truth labels from three annotators (the authors) provided
853 proxy AMCEs. Grid search minimised mean JSD on Qwen2.5-72B. Transferability was verified
854 on a disjoint 50-scenario MultiTP English subset: performance within 1.2% relative JSD of oracle-
855 tuned configuration. T_{cat} values reflect empirical logit magnitude distributions: Species and Gender
856 produce systematically larger gaps, requiring higher temperatures. Because this tuning pool is syn-
857 thetic and author-annotated, residual bias is possible; we therefore report broad robustness sweeps
858 (Appendix A10, Appendix A17) to show conclusions are stable beyond one tuned point.

859 A10.3 Temperature Sensitivity

860 We sweep the two temperature families on the (USA, DEU, JPN) subset of MultiTP using
861 Llama-3.1-70B. Sweep A varies $T_{\text{dec}} \in \{0.10, 0.25, 0.50, 0.75, 1.0, 2.0\}$; Sweep B replaces the
862 per-category temperatures by a single uniform $T_{\text{cat}} \in \{0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0\}$; Sweep C
863 varies the “Others” bucket $T_{\text{cat}} \in \{0.75, 1.0, 1.5, 2.0, 3.0, 4.0\}$ while holding $T_{\text{cat}}[\text{Species}] = 4.0$,
864 $T_{\text{cat}}[\text{Gender}] = 3.5$. All three sweeps yield JSD spans below 0.010 (0.0074, 0.0073, 0.0077 respec-
865 tively), and in every sweep the DISCA default (†) is strictly *more conservative* than the empirically

best value ($\Delta\text{JSD} \in \{-0.0061, -0.0021, -0.0037\}$). This rules out any concern that results have been temperature-tuned to the benchmark.

Table 21: Temperature sensitivity. Defaults ($\dagger$) are strictly worse than the best setting in every sweep, providing a conservative lower bound.

Sweep	Setting	JSD $\downarrow$	Pearson r $\uparrow$	ΔJSD
A: T_{dec}	0.10	0.0502	0.662	+0.0013
	0.25	0.0491	0.658	+0.0002
	0.50 $\dagger$	0.0489	0.630	-
	0.75	0.0476	0.636	-0.0013
	1.00	0.0466	0.641	-0.0023
	2.00 (best)	0.0428	0.627	-0.0061
	JSD span	0.0074		
B: Uniform T_{cat}	0.5	0.0521	0.642	+0.0052
	1.0	0.0492	0.673	+0.0023
	1.5 $\dagger$	0.0469	0.700	-
	2.0	0.0461	0.707	-0.0008
	3.0	0.0454	0.707	-0.0015
	4.0	0.0451	0.701	-0.0018
	6.0 (best)	0.0448	0.684	-0.0021
	JSD span	0.0073		
C: Others T_{cat}	0.75	0.0526	0.566	+0.0039
	1.00	0.0512	0.589	+0.0025
	1.50 $\dagger$	0.0487	0.634	-
	2.00	0.0475	0.660	-0.0012
	3.00	0.0458	0.693	-0.0029
	4.00 (best)	0.0450	0.706	-0.0037
	JSD span	0.0077		

The monotonic JSD reduction with larger T_{dec} reflects the role of T_{dec} in undoing RLHF logit compression: higher values widen the effective token-probability landscape so that IS perturbations exert larger directional influence. However, r peaks near $T_{\text{dec}}=1.0$ and declines at 2.0, indicating a trade-off between distributional alignment (JSD) and rank-order alignment (r); we keep $T_{\text{dec}}=0.5$ as a deliberately conservative operating point.

A10.4 Extended Hyperparameter Sensitivity

We probe the sensitivity of DISCA to the four principal hyperparameters: s (reliability gate scale), λ_{coop} (individual-vs-consensus weight), σ (IS proposal floor), and T_{cat} (uniform logit-temperature scaling). For each axis we sweep five values on a three-country panel (USA, VNM, DEU) and hold the other three axes at their defaults; the s axis directly probes how the choice of reliability scale in Eq. 5 affects downstream MIS.

Table 22: Hyperparameter sensitivity ranges on the three-country panel (Phi-4, $n=250/\text{country}$). Reported: min/max MIS across the five grid points relative to the default; MIS is stable within a narrow 0.0037–0.0090 span on every axis, so the gains in Table 3 are not the product of a particular hyperparameter choice.

Axis	Default	Grid	Min MIS	Max MIS	Δ
s	0.04	{0.01, 0.02, 0.04, 0.08, 0.16}	0.4091	0.4164	0.0073
λ_{coop}	0.70	{0.30, 0.50, 0.70, 0.85, 0.95}	0.4085	0.4173	0.0088
σ	0.30	{0.10, 0.20, 0.30, 0.45, 0.60}	0.4084	0.4122	0.0037
T_{cat} scale	1.0 $\times$	{0.33 $\times$, 0.67 $\times$, 1.0 $\times$, 1.33 $\times$, 1.67 $\times$ }	0.4060	0.4149	0.0090

879 **Why these four axes.** s controls the steepness of the reliability gate (Eq. 5); λ_{coop} trades off
880 individual-persona PT utility against the consensus utility; σ floors the IS proposal scale so the
881 $N=4$ empirical std cannot collapse to zero; and T_{cat} scales the whole family of per-category deci-
882 sion temperatures (preserving relative balance)-a global sharpening / flattening knob on the decision
883 logits. Per-category temperatures are stress-tested separately in the temperature sensitivity sweep
884 above.

885 **Persona count, IS budget, and per-category vs. global T_{cat} .** We additionally swept three oper-
886 ational knobs on the same USA/VNM/DEU panel. **Persona count** ($N \in \{2, 3, 4, 5, 6\}$, Ta-
887 ble 23): $N=4$ is optimal at macro MIS 0.4051, with degradation at both smaller and larger pan-
888 els (0.4252 at $N=2$, 0.4179 at $N=6$), supporting the four-persona design. **IS budget** ($K_{\text{half}} \in$
889 $\{8, 16, 32, 64, 128, 192\}$, Table 24): performance is flat across an order of magnitude (best 0.4092
890 at $K=16$; 0.4103 at $K=64$), indicating the default $K_{\text{half}}=64$ is in the stable region rather than over-
891 tuned. **Global vs. per-category T_{cat}** (Table 25): the per-category default outperforms *every* global
892 schedule we tested (0.4098 vs. all seven global values in $\{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$, which land
893 in 0.4119–0.4138). The advantage is small in absolute terms but consistent.

Table 23: Persona-count sweep on the USA/VNM/DEU panel (Phi-4, $n=250$ /country). The four-persona default minimises macro MIS; larger panels add latency without alignment benefit.

N	Macro MIS ↓	Std (across countries)	Flip rate	sec/scenario
2	0.4252	0.1045	0.275	0.089
3	0.4194	0.0945	0.273	0.100
4	0.4051	0.0829	0.276	0.111
5	0.4107	0.0745	0.276	0.120
6	0.4179	0.0919	0.245	0.130

Table 24: IS-budget sweep (K_{half} samples per pass; total per scenario is $K_{\text{total}}=2K_{\text{half}}$). Macro MIS is flat across an order of magnitude. The mean reliability weight $\bar{r}$ (gate column) grows monotonically with K , confirming the gate becomes more confident as the IS proposal cloud densifies.

K_{half}	K_{total}	Macro MIS ↓	Flip rate	$\bar{r}$ (gate)	sec/scen
8	16	0.4114	0.284	0.684	0.121
16	32	0.4092	0.271	0.736	0.114
32	64	0.4104	0.278	0.766	0.114
64	128	0.4103	0.280	0.793	0.113
128	256	0.4147	0.274	0.814	0.116
192	384	0.4126	0.273	0.827	0.112

Table 25: Global vs. per-category T_{cat} on the USA/VNM/DEU panel. The per-category default (decision 4.0, social 3.5, non-social 1.5) outperforms every global value we tested. The gap is small (≤ 0.004) but uniformly directional, supporting category-specific decision temperatures over a single shared scalar.

T_{cat} schedule	Macro MIS ↓	Std	Flip rate
per-category default (4.0 / 3.5 / 1.5)	0.4098	0.0881	0.278
global $T_{\text{cat}} = 1.00$	0.4120	0.0921	0.292
global $T_{\text{cat}} = 1.50$	0.4119	0.0838	0.304
global $T_{\text{cat}} = 2.00$	0.4138	0.0778	0.301
global $T_{\text{cat}} = 2.50$	0.4134	0.0774	0.296
global $T_{\text{cat}} = 3.00$	0.4127	0.0749	0.303
global $T_{\text{cat}} = 3.50$	0.4138	0.0689	0.306
global $T_{\text{cat}} = 4.00$	0.4128	0.0734	0.302

A11 Scenario-Slice Release and Category-Cap Sensitivity

Section 4.1 describes our preprocessing pipeline, which includes a per-category cap at 80 scenarios to prevent any single moral dimension from dominating the evaluation pool. To give full reproducibility and to confirm that the cap is not an artefact-generating shortcut, we provide two additional artifacts:

- **Exact scenario-slice release.** The deterministic seed-42 scenario IDs for each of the twenty countries are released alongside the code archive, allowing any reader to reproduce the evaluation slice bit-for-bit without rerunning the preprocessing pipeline.
- **No-cap sensitivity run.** We re-run Phi-4 DISCA on the twenty-country grid with the per-category cap disabled. This run executes 440 scenarios per country (vs. the capped slice) and yields macro MIS 0.4051, macro JSD 0.0662, and mean Pearson r 0.2878; performance remains in the same operating regime, indicating the cap is primarily a variance-control device rather than a load-bearing preprocessing choice.

Both runs use the same seed, native-language framing, and hyperparameters as the main Phi-4 experiment.

A12 Theoretical Grounding

DISCA sits at the intersection of three theoretical traditions, each supplying one ingredient that the others lack. *Control as probabilistic inference* [Levine, 2018, Todorov, 2009] recasts optimal decision-making as posterior inference over an intention distribution; path-integral steering [Williams et al., 2018] instantiates that view with softmax-weighted perturbations that provably approximate the KL-regularised optimal control under mild regularity conditions [Todorov, 2009]. This is what licences us to read our Eq. 4 as an inference step rather than an arbitrary scalar interpolation, and it motivates the Gaussian proposal around δ as the entropy-maximal default in the absence of stronger priors. *Prospect Theory* [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] then supplies the utility shape: the key empirical regularity-that human decision makers weight a loss of magnitude x roughly $\kappa \approx 2.25$ times more than a gain of the same magnitude-is exactly the asymmetry we want an LLM cultural controller to respect, because a correction that silences one demographic persona is worse than a correction that fails to help it. Recent evidence that prospect-theoretic biases emerge naturally in LLM decision-making under risk [Payne, 2025] strengthens the match: the asymmetric gain/loss structure we impose *mirrors* one the models themselves already exhibit, rather than introducing an alien objective. *Importance-sampling (IS) theory* [Elvira and Martino, 2021] finally supplies the variance-control machinery, and it is here that self-normalised IS is known to be biased at finite K ; the dual-pass construction we adopt is a light-weight bootstrap-style variance probe [Efron, 1979] that uses the same compute budget to detect instability rather than silently accept it, paired with an effective-sample-size guard grounded in design-effect theory [Kish, 1965]. We treat the dual-pass gate as a variance heuristic rather than a formal unbiasedness certificate; formal finite- K bounds would require stronger assumptions on the logit landscape than we are willing to claim (Appendix A14).

A13 Design Rationale: Why Each Component

This appendix collects the full justification for the three design choices previewed in §1 and §3: the four-persona WVS panel, the Prospect-Theory IS kernel, and the dual-pass reliability gate.

Why a panel, and why the World Values Survey? A single “prompt-as-country” encoder-e.g., “Answer as a Japanese person”—collapses a distribution to a point estimate and discards the very spread the correction stage needs. Cultural-value research has long argued against this framing: both Inglehart-Welzel’s modernisation theory [Inglehart and Welzel, 2005] and Schwartz’s refined theory of basic individual values [Schwartz et al., 2012] find within-country variance on any single value dimension comparable to between-country variance, with demographic cleavages-most prominently age cohort-driving the largest slice. The Moral Machine analysis confirms this specifically for trolley preferences: generational differences in how respondents weight youth, social status, and

species cut across every country cluster [Awad et al., 2018, Kleiman-Weiner et al., 2017]. WVS Wave 7 [Haerpfer et al., 2022] is the largest public instrument that reports *individual-level* micro-data with the relevant age, gender, and value coverage, making it the natural substrate for data-driven personas rather than author-crafted stereotypes—a well-documented failure mode of prompt-based persona work [Deshpande et al., 2023].

Why $N=4$ specifically? Four is the smallest panel that simultaneously covers the three demographic axes documented to carry most within-country variance—young (<36), middle-aged (36–55), older (>55)—plus a country-wide aggregate that anchors the ensemble against cohort-specific sampling noise. Three personas would sacrifice the population anchor and leave the consensus $\bar{\delta}$ vulnerable to a single outlier cohort; five or more would duplicate coverage of the same axis without adding a new source of variance (WVS does not resolve finer cohorts reliably for every country). The cohort structure matches the statistical-survey tradition of stratifying by age before any country-level estimate [Kish, 1965]. Empirically, the ablation in §6 confirms the ensemble is load-bearing: removing it costs -0.183 Pearson r and $+0.083$ MIS.

Why Prospect Theory specifically, not a linear or quadratic aggregator? Three independent considerations pick out the Kahneman-Tversky value function [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] over simpler alternatives. (1) *Asymmetric harm*. In a cultural panel, a correction that improves three personas while making one worse is not equivalent to one that helps all four equally: the former silences a demographic group the survey says exists. Linear averaging is oblivious to this asymmetry; quadratic aggregation treats $+x$ and $-x$ as equally valuable and costly. Loss aversion with $\kappa \approx 2.25$ is the textbook object for encoding “harm is worse than equal-sized help”. (2) *Diminishing sensitivity matches the scale of the cultural signal*. Gain curvature $\alpha = 0.88 < 1$ compresses large persona gains relative to small ones, preventing one outlier persona (perhaps from sparse WVS coverage) from dominating the utility [Tversky and Kahneman, 1992]. (3) *Empirical compatibility with the base model*. Recent work shows that LLMs exhibit prospect-theoretic biases in decision-making under risk, with loss-aversion coefficients in the same neighbourhood as human respondents [Payne, 2025]. While we do not claim that logit-gap changes in an LLM are psychologically equivalent to monetary gains and losses in human subjects—the PT value function is used here as a *controller aggregation kernel*, not as a cognitive model of the LLM—the empirical compatibility means we impose a utility shape that does not fight the model’s own implicit asymmetries. Auxiliary runs with linear reward kernels and WVS-weighted linear kernels on the same pipeline confirm that the PT shape outperforms symmetric alternatives, and the robustness sweeps in Appendix A10 show that the method’s conclusions are stable across three logit-shift kernels, so the specific $\kappa = 2.25$ is a well-motivated default rather than a sensitive tuning knob.

Why importance sampling at all? The cultural correction we seek is the argmax of a utility that values each persona’s alignment and penalises corrections that make any of them worse [Levine, 2018]. Closed-form optimisation is unavailable because the utility is non-convex (PT’s kink at zero, the ℓ_1 -style gain definition in Eq. 13), so we approximate the optimal control distribution with a path-integral-style softmax over perturbations [Williams et al., 2018, Todorov, 2009]. The Gaussian proposal $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ plays the role of an entropy-maximal default, and the softmax weights $w_k \propto \exp(U_{\text{total}}/\eta)$ implement the KL-regularised Boltzmann posterior balancing utility against deviation from the consensus. This control-as-inference reading licenses Eq. 4 as an approximate Bayesian update rather than arbitrary scalar interpolation.

Why dual-pass, rather than more samples or a hard threshold? Self-normalised IS is a biased estimator at finite sample size [Elvira and Martino, 2021]: the bias shrinks as $O(1/K)$ but does not reveal itself to the caller, so “more samples” hides instability rather than flagging it. Two independent replicates of size K_{half} using the same total budget $K = 2K_{\text{half}}$ trade a small increase in per-pass variance for a *direct, scenario-level probe* of that instability: the squared gap $(\delta_1^* - \delta_2^*)^2$ is a one-dimensional bootstrap-style variance estimator [Efron, 1979] whose small values license the correction and whose large values veto it automatically, without a hand-tuned threshold. The exponential map $r = \exp(-(\delta_1^* - \delta_2^*)^2/s)$ is preferred over a hard indicator because it preserves a single differentiable code path and its decay matches the shape of a Gaussian likelihood ratio under independent noise. An effective-sample-size guard [Kish, 1965] on each pass ($N_{\text{eff}}/K_{\text{half}} < \rho_{\text{eff}}$) zeros the offset entirely when the weights collapse, providing a second, coarser safety net below the smooth reliability gate. We emphasise that this construction is a variance heuristic, not a formal

998 unbiasedness certificate; finite- K bounds would require stronger assumptions on the logit landscape
 999 than we are willing to claim (Appendix A14).

1000 **A14 PT-IS, Sampling, and Dual-Pass Reliability**

1001 **A14.1 Gain definitions**

1002 For candidate $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$, the per-persona and consensus gains are:

$$g_{i,k} = |\delta_{\text{base}} - \delta_i| - |\tilde{\delta}_k - \delta_i|, \quad (13)$$

$$g_{\text{cons},k} = |\delta_{\text{base}} - \bar{\delta}| - |\tilde{\delta}_k - \bar{\delta}|. \quad (14)$$

1003 $g_{i,k} > 0$ means the candidate is closer to persona i than the base model; $g_{\text{cons},k}$ is the analogous
 1004 quantity for the persona consensus. These enter Eq. 4 after σ -normalisation. The IS update for each
 1005 pass is $\delta_m^* = \sum_k w_k^{(m)} \epsilon_k$ with $w_k^{(m)} = \text{softmax}(U_{\text{total}}(\epsilon_k)/\eta)$ and temperature $\eta = 0.5$. An ESS
 1006 guard zeros δ_m^* when the Kish effective sample size $N_{\text{eff}}/K_{\text{half}}$ falls below $\rho_{\text{eff}} = 0.1$.

1007 **A14.2 Utility shape and the omitted explicit KL-on- ϵ term**

1008 The scalar decision gap δ acts as a one-dimensional state and the perturbations ϵ_k as control inputs
 1009 at a single horizon step. Under the path-integral / free-energy view [Williams et al., 2018], softmax
 1010 weights $w_k \propto \exp(U_{\text{total}}/\eta)$ can be read as targeting a Boltzmann distribution over candidate pertur-
 1011 bations. **Our released code does not add** a separate quadratic penalty $\propto \epsilon_k^2/(2\sigma^2)$ to U_{total} in Eq. 4:
 1012 the released controller matches it exactly, and the Gaussian proposal $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ already limits
 1013 how far $\tilde{\delta}_k$ can move from $\bar{\delta}$. An optional explicit KL-on-control term remains a straightforward
 1014 extension if future work needs sharper tail control on $|\epsilon_k|$.

1015 **A14.3 Comparison with Simpler Alternatives**

1016 Temperature scaling uniformly stretches the logit gap without accounting for direction or magnitude
 1017 of persona rewards. A fixed margin offset ignores the utility landscape, treating all perturbations as
 1018 equally valuable. PT-weighted IS performs an *adaptive, scenario-specific* correction via nonlinear
 1019 importance weighting. Auxiliary runs with deterministic consensus-only shifts and WVS-linear
 1020 reward kernels underperform the Prospect-Theory kernel on the same pipeline; we omit utility-shape
 1021 rows from Table 5 for clarity.

1022 **A14.4 Why split K into two passes?**

1023 Self-normalised IS is biased at finite sample size; two independent halves with the *same* total
 1024 budget trade per-pass variance for a direct probe of sampling instability. The reliability map
 1025 $r = \exp(-(\delta_1^* - \delta_2^*)^2/s)$ is intentionally smooth-unlike a hard veto it preserves a single code path
 1026 and down-weights only when the two estimates contradict each other strongly. It should be read as
 1027 a variance heuristic, not as a statistical guarantee of unbiasedness; formal finite- K bounds would
 1028 require stronger assumptions on the logit landscape than we are willing to claim here.

1029 **A14.5 Positional Debiasing: Implementation Detail**

1030 The two-pass debiasing of §3.4 is implemented as a text-level rewrite of the scenario *after* native-
 1031 language framing. To avoid the trivial bug of double-substitution (replacing LEFT $\rightarrow$ RIGHT im-
 1032 mediately followed by RIGHT $\rightarrow$ LEFT would leave every label as LEFT), we use a placeholder-
 1033 based three-step swap, applied independently for the lane-label pair and the group-label pair:

- 1034 1. Replace every occurrence of the left label with a unique placeholder token (a non-printing
 1035 sentinel).
- 1036 2. Replace every occurrence of the right label with the left label.
- 1037 3. Replace the placeholder with the right label.

1038 The same procedure is applied to Group A $\leftrightarrow$ Group B in the native script. Logit gaps ($\delta_{\text{base}}, \delta_i$)
1039 from the two orders are combined *before* any nonlinear step, $\delta \leftarrow (\delta^{(\text{orig})} - \delta^{(\text{swap})})/2$, matching the
1040 released controller (debias first, then PT-IS and dual-pass merge). This guarantees cancellation of
1041 additive positional bias in logit space; the final p_{spare} applies $\sigma(\cdot)$ only after that correction.

1042 **A15 Persona Construction Details**

1043 This appendix specifies how the four cultural personas per country are constructed and provides
1044 empirical support that the construction is grounded in human survey data rather than authored ad hoc.
1045 §A15.1 reproduces two personas verbatim from `generate_wvs_persona()`; §A15.2 documents
1046 the WVS Wave 7 processing pipeline; §A15.3 links the ten WVS cultural dimensions to the six
1047 MultiTP moral attributes both thematically and empirically; and §A15.4 shows that DISCA is robust
1048 to the choice of the fourth (anchor) persona.

1049 **A15.1 Persona Prompts**

1050 Each country yields four personas: three age-cohort personas (*young, middle, older*) generated from
1051 the country-and-cohort-specific WVS-7 means, plus a population-wide aggregate that serves as a
1052 demographic anchor. Below we provide the full prompt sets for the United States and Vietnam,
1053 reproduced verbatim from the WVS-driven persona generation pipeline with the aggregate fourth
1054 persona setting.

1055 **United States (all four personas, verbatim)**

1056 **Young cohort.**

1057 *You are a young adult from the United States, in your 20s and early 30s. Your worldview*
1058 *is shaped by the cultural values prevalent in your community. On matters of faith you are*
1059 *moderately religious. On raising children you are firmly oriented toward independence and*
1060 *imagination. On contested moral choices you are morally conservative on contested issues.*
1061 *In your dealings with strangers you have a guarded attitude toward strangers. Civically*
1062 *you are an active political participant who signs petitions, joins boycotts and takes part in*
1063 *lawful demonstrations. You are moderately proud of your country. Overall you are rather*
1064 *happy with your life. On the role of women in society you are moderately egalitarian*
1065 *on gender roles. In what you prioritise in life you are leaning materialist, prioritising*
1066 *economic and physical security. Toward people unlike yourself you are highly tolerant of*
1067 *outgroups such as immigrants, minorities and people with different lifestyles. When you*
1068 *face a moral dilemma, you weigh the choices through this set of values and answer in a*
1069 *way that is consistent with the worldview above.*

1070 **Middle-aged cohort.**

1071 *You are a middle-aged adult from the United States, in your 40s or 50s. Your worldview*
1072 *is shaped by the cultural values prevalent in your community. On matters of faith you are*
1073 *moderately religious. On raising children you are firmly oriented toward independence*
1074 *and imagination. On contested moral choices you are strictly opposed to such contested*
1075 *moral acts. In your dealings with strangers you have a guarded attitude toward strangers.*
1076 *Civically you are an active political participant who signs petitions, joins boycotts and*
1077 *takes part in lawful demonstrations. You are moderately proud of your country. Overall*
1078 *you are rather happy with your life. On the role of women in society you are moderately*
1079 *egalitarian on gender roles. In what you prioritise in life you are leaning materialist,*
1080 *prioritising economic and physical security. Toward people unlike yourself you are highly*
1081 *tolerant of outgroups such as immigrants, minorities and people with different lifestyles.*
1082 *When you face a moral dilemma, you weigh the choices through this set of values and*
1083 *answer in a way that is consistent with the worldview above.*

1084 **Older cohort.**

1085 *You are a senior citizen from the United States, over 60 years old. Your worldview is*
1086 *shaped by the cultural values prevalent in your community. On matters of faith you are*
1087 *moderately religious. On raising children you are firmly oriented toward independence*
1088 *and imagination. On contested moral choices you are strictly opposed to such contested*

1089 *moral acts. In your dealings with strangers you have a guarded attitude toward strangers.*
 1090 *Civically you are an active political participant who signs petitions, joins boycotts and*
 1091 *takes part in lawful demonstrations. You are intensely proud of your country. Overall*
 1092 *you are rather happy with your life. On the role of women in society you are moderately*
 1093 *egalitarian on gender roles. In what you prioritise in life you are leaning materialist,*
 1094 *prioritising economic and physical security. Toward people unlike yourself you are highly*
 1095 *tolerant of outgroups such as immigrants, minorities and people with different lifestyles.*
 1096 *When you face a moral dilemma, you weigh the choices through this set of values and*
 1097 *answer in a way that is consistent with the worldview above.*

1098 **Aggregate cohort (all ages).**

1099 *You are a adult citizen from the United States. Your worldview is shaped by the cultural*
 1100 *values prevalent in your community. On matters of faith you are moderately religious.*
 1101 *On raising children you are firmly oriented toward independence and imagination. On*
 1102 *contested moral choices you are strictly opposed to such contested moral acts. In your*
 1103 *dealings with strangers you have a guarded attitude toward strangers. Civically you are*
 1104 *an active political participant who signs petitions, joins boycotts and takes part in lawful*
 1105 *demonstrations. You are moderately proud of your country. Overall you are rather happy*
 1106 *with your life. On the role of women in society you are moderately egalitarian on gender*
 1107 *roles. In what you prioritise in life you are leaning materialist, prioritising economic and*
 1108 *physical security. Toward people unlike yourself you are highly tolerant of outgroups such*
 1109 *as immigrants, minorities and people with different lifestyles. When you face a moral*
 1110 *dilemma, you weigh the choices through this set of values and answer in a way that is*
 1111 *consistent with the worldview above.*

1112 **Vietnam (all four personas, Vietnamese, verbatim)**

1113 **Young cohort.**

1114 *Bạn là một thanh niên đến từ Việt Nam, ở độ tuổi 20 đến đầu 30. Thế giới quan của bạn*
 1115 *được định hình bởi các giá trị văn hóa phổ biến trong cộng đồng của bạn. Về vấn đề đức*
 1116 *tin, bạn khá thể tục. Trong việc nuôi dạy con cái, bạn nghiêng về sự vắng lời và đức tin tôn*
 1117 *giáo. Về các lựa chọn đạo đức gây tranh cãi, bạn bảo thủ về mặt đạo đức trong các vấn đề*
 1118 *gây tranh cãi. Trong giao tiếp với người lạ, bạn có sự ngờ vực sâu sắc đối với người khác.*
 1119 *Về mặt công dân, bạn là một người tham gia chính trị thụ động. Bạn rất tự hào về đất nước*
 1120 *của bạn. Nhìn chung, bạn rất hài lòng với cuộc sống của bạn. Về vai trò của phụ nữ trong*
 1121 *xã hội, bạn khá truyền thống về vai trò giới. Về những gì bạn ưu tiên trong cuộc sống, bạn*
 1122 *có xu hướng hậu vật chất. Đối với những người khác bạn, bạn có phần thiếu khoan dung*
 1123 *với các nhóm bên ngoài. Khi bạn đối mặt với một tình huống khó xử về đạo đức, bạn cân*
 1124 *nhắc các lựa chọn thông qua tập hợp giá trị này và trả lời theo cách phù hợp với thế giới*
 1125 *quan đã nêu ở trên.*

1126 **Middle-aged cohort.**

1127 *Bạn là một người trung niên đến từ Việt Nam, ở độ tuổi 40 hoặc 50. Thế giới quan của bạn*
 1128 *được định hình bởi các giá trị văn hóa phổ biến trong cộng đồng của bạn. Về vấn đề đức*
 1129 *tin, bạn khá thể tục. Trong việc nuôi dạy con cái, bạn nghiêng về sự vắng lời và đức tin tôn*
 1130 *giáo. Về các lựa chọn đạo đức gây tranh cãi, bạn bảo thủ về mặt đạo đức trong các vấn đề*
 1131 *gây tranh cãi. Trong giao tiếp với người lạ, bạn có thái độ dè dặt với người lạ. Về mặt công*
 1132 *dân, bạn là một người thỉnh thoảng tham gia chính trị. Bạn rất tự hào về đất nước của bạn.*
 1133 *Nhìn chung, bạn rất hài lòng với cuộc sống của bạn. Về vai trò của phụ nữ trong xã hội,*
 1134 *bạn khá truyền thống về vai trò giới. Về những gì bạn ưu tiên trong cuộc sống, bạn có xu*
 1135 *hướng hậu vật chất. Đối với những người khác bạn, bạn có phần thiếu khoan dung với các*
 1136 *nhóm bên ngoài. Khi bạn đối mặt với một tình huống khó xử về đạo đức, bạn cân nhắc các*
 1137 *lựa chọn thông qua tập hợp giá trị này và trả lời theo cách phù hợp với thế giới quan đã*
 1138 *nêu ở trên.*

1139 **Older cohort.**

1140 *Bạn là một người cao tuổi đến từ Việt Nam, trên 60 tuổi. Thế giới quan của bạn được định*
 1141 *hình bởi các giá trị văn hóa phổ biến trong cộng đồng của bạn. Về vấn đề đức tin, bạn có*
 1142 *niềm tin tôn giáo ở mức trung bình. Trong việc nuôi dạy con cái, bạn nghiêng về sự vắng*
 1143 *lời và đức tin tôn giáo. Về các lựa chọn đạo đức gây tranh cãi, bạn bảo thủ về mặt đạo đức*
 1144 *trong các vấn đề gây tranh cãi. Trong giao tiếp với người lạ, bạn có thái độ dè dặt với người*

lạ. Về mặt công dân, bạn là một người thỉnh thoảng tham gia chính trị. Bạn rất tự hào về đất nước của bạn. Nhìn chung, bạn rất hài lòng với cuộc sống của bạn. Về vai trò của phụ nữ trong xã hội, bạn khá truyền thống về vai trò giới. Về những gì bạn ưu tiên trong cuộc sống, bạn có xu hướng hậu vật chất. Đối với những người khác bạn, bạn rất thiếu khoan dung với các nhóm bên ngoài. Khi bạn đối mặt với một tình huống khó xử về đạo đức, bạn cân nhắc các lựa chọn thông qua tập hợp giá trị này và trả lời theo cách phù hợp với thể giới quan đã nêu ở trên.

Aggregate cohort (all ages).

Bạn là một công dân trưởng thành đến từ Việt Nam. Thể giới quan của bạn được định hình bởi các giá trị văn hóa phổ biến trong cộng đồng của bạn. Về vấn đề đức tin, bạn khá thể tục. Trong việc nuôi dạy con cái, bạn nghiêng về sự vâng lời và đức tin tôn giáo. Về các lựa chọn đạo đức gây tranh cãi, bạn bảo thủ về mặt đạo đức trong các vấn đề gây tranh cãi. Trong giao tiếp với người lạ, bạn có thái độ dè dặt với người lạ. Về mặt công dân, bạn là một người thỉnh thoảng tham gia chính trị. Bạn rất tự hào về đất nước của bạn. Nhìn chung, bạn rất hài lòng với cuộc sống của bạn. Về vai trò của phụ nữ trong xã hội, bạn khá truyền thống về vai trò giới. Về những gì bạn ưu tiên trong cuộc sống, bạn có xu hướng hậu vật chất. Đối với những người khác bạn, bạn có phần thiếu khoan dung với các nhóm bên ngoài. Khi bạn đối mặt với một tình huống khó xử về đạo đức, bạn cân nhắc các lựa chọn thông qua tập hợp giá trị này và trả lời theo cách phù hợp với thể giới quan đã nêu ở trên.

Aggregate (fourth) persona. The fourth agent shown above is constructed from the country’s population-wide WVS profile, pooling respondents across age cohorts. It serves as a demographic anchor: by suppressing cohort-specific noise it ensures that the ensemble’s consensus target δ reflects the broadest available empirical signal rather than a single generational viewpoint. §A15.4 contrasts this choice with a country-invariant utilitarian anchor and shows that DISCA’s macro-MIS is insensitive to the substitution.

Faithfulness disclaimer. The descriptors above are read directly from the country’s WVS-7 means through deterministic quartile cuts (Table 26); they are not hand-selected to match cultural stereotypes. As a consequence, individual cohorts can present empirically grounded but locally counter-intuitive combinations (for example, a U.S. young-adult that is *leaning materialist* and *morally conservative on contested issues*), reflecting actual WVS-7 distributions rather than authorial bias.

A15.2 WVS Data Processing Pipeline

We process WVS Wave 7 individual-level microdata following the ten-variable cultural-value scheme of Greco et al. [2026]. The pipeline is fully deterministic: given the inverted WVS-7 CSV and the country code, it returns the four personas used at inference time without any tunable hyperparameters.

1. **Variable extraction:** For each respondent, we extract 10 value dimensions from WVS items (Table 26). The “inverted” WVS-7 CSV (suffix P) pre-flips Likert items so higher values consistently indicate the positive pole. Items without the P suffix (Q152, Q153, Q177–Q182) use original WVS coding.
2. **Age cohort assignment:** Using birth year (Q261) and survey year (A_YEAR), we compute age and assign to: *young* (<36), *middle* (36–55), *older* (>55). Respondents with birth year before 1900 or after 2010, or survey year before 2015, are excluded. Negative WVS codes (−1, −2, −4, −5: refusal / don’t know) are dropped; zero is retained for binary 0/1 items.
3. **Cohort aggregation:** For each country and age cohort, each dimension’s score is the unweighted mean of all valid (respondent, item) values associated with that dimension - i.e., for multi-item dimensions (e.g., Q177–Q182 for moral acceptability) every respondent contributes one value per item to a single pooled mean. For single-item dimensions this reduces to the standard across-respondent mean.
4. **Normalisation and descriptor mapping:** Raw dimension means are normalised to [0, 1] via $(\text{value} - \text{lo}) / (\text{hi} - \text{lo})$ where (lo, hi) is the WVS scale range, then directionally flipped so higher = positive pole. Four-level descriptors are assigned by quartile cuts: ≥ 0.75 (strong positive), $\geq 0.50, \geq 0.25, < 0.25$ (strong negative). See Table 26.

1197 5. **Fallback:** For countries with insufficient WVS coverage (e.g., Saudi Arabia), we use man-
1198 ually authored native-language personas grounded in area-studies literature, covering the
1199 same demographic structure (3 age cohorts + 1 population-wide aggregate).

Table 26: The 10 WVS cultural-value dimensions used for persona generation, following Greco et al. [2026]. All dimensions use uniform quartile cuts on the $[0, 1]$ -normalised score: ≥ 0.75 (descriptor 1, strongest positive pole), ≥ 0.50 (2), ≥ 0.25 (3), < 0.25 (4, strongest negative pole). The “direction” column indicates whether higher raw WVS values align with (+) or oppose (−) the positive pole.

Dimension	WVS items	Range	Dir.	Descriptors (≥ 0.75 / ≥ 0.50 / ≥ 0.25 / < 0.25)
Religiosity	Q6P	1–4	+	deeply / moderately religious / somewhat / highly secular
Child-rearing	Q17P	0–1	−	independence-imagination $\leftrightarrow$ obedience-faith
Moral accept.	Q177–Q182	1–10	+	very permissive $\leftrightarrow$ strictly opposed
Social trust	Q57P	1–2	+	very high trust $\leftrightarrow$ deep distrust
Polit. particip.	Q199P, Q200P	1–3	+	active participant $\leftrightarrow$ non-participant
National pride	Q254P	1–4	+	very proud $\leftrightarrow$ not proud
Happiness	Q46P	1–4	+	very happy $\leftrightarrow$ not happy
Gender equality	Q29P, Q30P, Q31P, Q33P	1–4	−	strongly egalitarian $\leftrightarrow$ traditional roles
Materialism	Q152, Q153	1–3	+	post-materialist $\leftrightarrow$ materialist
Tolerance	Q19P–Q23P	0–1	−	very tolerant $\leftrightarrow$ intolerant

1200 A15.3 WVS-to-Trolley Dimension Linkage

1201 The linkage between the ten WVS cultural dimensions and the six MultiTP moral dimensions oper-
1202 ates through two complementary mechanisms.

1203 **Direct thematic overlap.** WVS *gender equality* maps onto the Gender dimension-societies scor-
1204 ing higher exhibit weaker female-sparing preferences [Awad et al., 2018]; *religiosity* correlates with
1205 Species through religious anthropocentrism and human exceptionalism; and *moral acceptability* re-
1206 lates to tolerance for trade-offs along the Fitness and Social Value dimensions.

1207 **Indirect modulation.** Dimensions such as *social trust*, *child-rearing values*, and *materialism ori-*
1208 *entation* reshape the moral *frame* the persona adopts: post-materialist personas favour utilitarian
1209 calculi, while obedience-oriented child-rearing values up-weight hierarchical social roles (affecting
1210 Age and Social Value). Importantly, DISCA does not require an exact causal mapping between
1211 WVS and MultiTP dimensions; the ten-dimensional persona ensemble produces diversity in logit-
1212 space gaps, and the importance-sampling stage selects the correction that balances collective utility.

1213 **Empirical validation of the WVS→AMCE pathway.** Table 27 regresses each of the six MultiTP
1214 human AMCE dimensions on the 10 WVS cultural features across the 20-country panel (standard-
1215 ised OLS). The 10-feature WVS profile explains $R^2 \in [0.55, 0.69]$ of human AMCE variance, with
1216 $|\beta| > 0.3$ in 31 of 60 cells. This is direct evidence that WVS-grounded persona construction is not
1217 arbitrary: the cultural axes our personas inherit correlate strongly with the moral preferences DISCA
1218 must reproduce.

Table 27: WVS features as predictors of human AMCE dimensions. OLS with standardised pre-
dictors. Coefficients are standardised (β); bold = $|\beta| > 0.3$. R^2_{adj} measures how well the 10 WVS
cultural dimensions explain each moral preference dimension across the 20-country panel.

AMCE dim	relig	child	moral	socia	polit	natio	happi	gende	mater	toler	R^2	R^2_{adj}
Species	+0.47	+0.12	-0.65	+0.60	+0.08	+0.16	-0.02	-0.74	-0.30	+0.40	0.57	0.09
Gender	+0.69	+0.37	-0.18	+0.41	-0.44	-0.39	+0.71	-0.81	-0.20	+0.06	0.66	0.29
Age	-0.80	-0.09	+0.61	-0.74	+0.17	+0.26	-0.27	+0.62	+0.49	-0.37	0.55	0.05
Fitness	+0.28	-0.23	-0.16	-0.12	-0.17	+0.43	+0.09	-1.46	-0.21	+0.53	0.60	0.15
Social Value	-0.27	-0.30	+0.40	-0.40	-0.37	+0.58	+0.17	-0.31	-0.12	-0.15	0.69	0.35
Utilitarianism	+1.11	+0.39	+0.07	+0.59	-0.21	-0.01	+0.13	-0.76	+0.42	+0.25	0.66	0.29

1219 Table 28 closes the loop with a causal leave-one-WVS-dim-out probe: for each WVS dimension
1220 d , we re-run DISCA with d removed from every persona and measure the change in per-MultiTP-

dim absolute error (positive cell = error rises when the dimension is removed = that dimension is load-bearing for the corresponding moral attribute). The three largest positive couplings are *moral acceptability* → Social Value (+3.97 pp), *social trust* → Social Value (+2.81 pp), and *child-rearing values* → Utilitarianism (+2.41 pp). Cells with the top-3 largest positive couplings per row are bolded; rows with fewer than three positive cells are bolded only on those positive cells. Negative cells are also informative: *tolerance of diversity* → Social Value (−7.76) and *gender equality* → Gender (−3.42) indicate that, for those moral attributes, the corresponding WVS descriptor biases the persona ensemble away from the human target, so removing the descriptor improves accuracy. Crucially, DISCA does not require manual feature selection to handle these noisy dimensions: the loss-averse importance-sampling stage (Eq. 4) automatically down-weights candidate perturbations that worsen any persona’s alignment, effectively acting as a built-in noise filter. When a WVS dimension introduces bias for a particular moral attribute, the PT-IS utility penalises the resulting correction asymmetrically ($\kappa=2.25\times$ for losses vs. gains), ensuring that noisy persona signals are suppressed rather than propagated. This self-correcting property is why the method remains robust despite imperfect WVS-to-trolley mappings—a design-level answer to the concern that the linkage is “indirect modulation” rather than causal.

Table 28: WVS-dim $\times$ MultiTP-dim causal impact matrix. Cells are the mean increase in per-dim AMCE error (pp) when the WVS dimension is dropped from every persona, macro-averaged across the country panel. **Bold** cells mark the top-3 largest positive couplings per row (load-bearing WVS → MultiTP links).

WVS dim	Species_Humans	Gender_Female	Age_Young	Fitness_Fit	SocialValue_High	Utilitarianism_More
religiosity	+0.66	+1.65	-1.66	+0.58	+1.02	-0.37
child rearing	-1.82	+1.29	+0.86	+1.30	-0.33	+2.41
moral acceptability	+1.13	-0.02	-1.56	-0.34	+3.97	-1.69
social trust	+0.11	+1.43	-0.85	+0.51	+2.81	-1.88
political participation	-1.24	-0.82	-0.63	-0.36	-0.16	+0.86
national pride	-0.37	-0.10	-0.28	+0.52	+1.19	+1.21
happiness	-0.23	-0.88	+0.13	-1.69	-0.11	+1.72
gender equality	+0.41	-3.42	-2.90	-1.11	-1.87	+0.62
materialism orientation	+1.35	-1.26	+0.90	+0.04	-0.36	+2.15
tolerance diversity	-0.76	-3.52	+1.95	-0.59	-7.76	+1.54

A15.4 Sensitivity to the Fourth Persona

Setup. §3.2 pairs three age-cohort personas with a population-wide *aggregate* WVS profile as the fourth agent. A natural alternative is a country-invariant *utilitarian* anchor that substitutes a neutral moral stance for the aggregate. To confirm that this choice is not load-bearing, we ran a head-to-head comparison of the two variants on the twenty-country Phi-4 grid.

Result. The utilitarian-anchor variant improves macro MIS from 0.4252 to 0.4049 ($\Delta = -0.0203$), a marginal gain that does not change any qualitative conclusion of the paper.

Decision. We retain the aggregate variant as the default to preserve strict country-grounding of every persona and to avoid importing a globally fixed moral prior into country-conditioned steering. The utilitarian variant remains available as an opt-in for practitioners who prefer a neutral anchor (`build_country_personas(..., fourth='utilitarian')`).

A16 AMCE Estimation Details

Uniform treatment of all dimensions. All six dimensions-Species, Gender, Age, Fitness, Social Value, and Utilitarianism—are computed identically. For each dimension, we compute the empirical mean of $p_{\text{spare}}(\mathbf{x})$ across all scenarios in that dimension (Eq. 6). For the five binary dimensions, each scenario presents a forced choice between two character groups differing on exactly one attribute. For Utilitarianism, scenarios vary the number of lives on each side; the binary indicator is whether the larger group (“More”) or the smaller group (“Less”) is spared. This uniform mean-based estimation is consistent with the `compute_ACME` method in the MultiTP codebase [Jin et al., 2025], which fits a no-intercept linear regression with a binary group indicator-mathematically equivalent to taking the mean of the saving probability for the preferred group.

Human AMCE scale. The MultiTP ground-truth AMCEs in the MultiTP-released country-specific AMCE table is on a $[-1, 1]$ scale. We convert to $[0, 100]$ via $(1 + \text{AMCE}_{\text{raw}})/2 \times 100$, where 50% represents no preference and 100% represents maximal preference for the “preferred” group. Model AMCEs are computed on the same scale.

Consistency. Awad et al. [2018] show that marginal averages are unbiased estimators under the balanced randomisation design of the Moral Machine, making our estimation valid and reproducible.

A17 Dataset Preprocessing and Sensitivity

Pipeline. Each MultiTP CSV is processed in five steps: (i) deduplication keeps only the canonical paraphrase (which_paraphrase=0); (ii) the Utilitarianism *quality filter* drops scenarios where the two groups have identical size *and* all characters belong to the *quality-attribute* role set {PREGNANT, WOMAN, LARGE WOMAN}, which would otherwise inject a non-numerosity signal into a dimension whose AMCE is by construction the count-difference effect; (iii) per-category capping retains at most 80 scenarios per dimension; (iv) under-represented categories are oversampled with replacement to a minimum of 36 scenarios; and (v) the resulting pool is shuffled with a fixed seed (42).

Reproducible left/right assignment. When the MultiTP paraphrase_choice field does not unambiguously determine which group is rendered on the left vs. right, we deterministically fall back to a SHA-256 hash of the tuple $(\text{sub}_1, \text{sub}_2, g_1, g_2)$ modulo 2, so the same scenario receives the same ordering across runs and machines. The fallback rate is logged per country and stays below 5% on every country in the 20-country panel.

Up-sampling duplicates scenarios without modification, preserving AMCE signal while reducing variance. Category capping prevents dimension dominance. Side balancing enables effective debiasing. To evaluate whether these choices alter the effective evaluation distribution, we compare *six* preprocessing configurations on 10 representative countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR, RUS, NGU) using Llama-3.1-70B.

Table 29: Per-country dataset sensitivity. JSD range = 0.0034 (aggregate) and < 0.010 in 9/10 countries; the only exception is RUS at 0.013, driven by augmentation sensitivity in the rare Species/Utilitarianism categories. D0 is retained as default.

Country	D0-Default	D1-NoAug	D2-Cap40	D3-Cap120	D4-NoFlip	D5-Strict
USA	0.0655	0.0674	0.0625	0.0644	0.0643	0.0684
DEU	0.0465	0.0515	0.0492	0.0467	0.0444	0.0484
CHN	0.0393	0.0363	0.0385	0.0383	0.0409	0.0398
JPN	0.0304	0.0291	0.0319	0.0296	0.0273	0.0320
BRA	0.0492	0.0551	0.0479	0.0510	0.0520	0.0480
VNM	0.0592	0.0574	0.0631	0.0585	0.0578	0.0626
GBR	0.0614	0.0667	0.0608	0.0618	0.0597	0.0633
KOR	0.0371	0.0354	0.0378	0.0357	0.0349	0.0382
RUS	0.0383	0.0489	0.0389	0.0374	0.0383	0.0357
NGU	0.0522	0.0568	0.0506	0.0525	0.0514	0.0538
Mean	0.0479	0.0505	0.0481	0.0476	0.0471	0.0490
Δ vs D0	-	+0.0025	+0.0002	-0.0003	-0.0008	+0.0011

D1-NoAug shows the largest degradation (+0.0025), justifying synthetic augmentation of the under-represented Species and Utilitarianism categories. D4-NoFlip is marginally lower than D0, indicating that group-side randomisation is conservative (it adds noise) rather than harmful; we retain it because it preserves positional-debiasing integrity. The 0.0034 aggregate range is well below the ± 0.004 bootstrap CI, so reported results are not confounded by preprocessing choices.

A18 Cross-Lingual Token Elicitation

Native-language prompting system. Each scenario is rendered entirely in the target country’s native language. The implementation includes full translations of: (i) scenario framing and context sentences (4 paraphrase variants per language), (ii) all 22 character types with singular/plural forms

1292 in each language, (iii) lane labels and group identifiers in native script, and (iv) closing questions
1293 with language-appropriate conjunctions and grammar.

1294 The prompt frame wraps the scenario in a native-language instruction that explicitly requests an En-
1295 glish answer token. Each frame follows the pattern: “[moral dilemma preamble in native language]
1296 \n {scenario} \n [instruction to answer A or B in English]”. This ensures moral reasoning occurs in
1297 the culturally native linguistic frame while maintaining a consistent decision interface.

1298 **Token verification.** Token IDs are verified per model at initialisation by encoding the upper-
1299 case strings "A" and "B" with `add_special_tokens=False` and asserting single-token mappings:
1300 Qwen2.5: A=22397, B=11572; Llama-3.1: A=31266, B=9268. Logits for these two positions are
1301 extracted from the last-position output tensor; no generation sampling is performed. The model uses
1302 its built-in `chat_template` for correct role formatting across model families.

1303 A19 Baseline Implementation Details

1304 A19.1 Activation Steering

1305 For each country, a steering vector is computed as the difference in mean residual-stream activations
1306 from 20 culturally contrastive prompt pairs (e.g., “Answer as someone from [country] who values
1307 [WVS-high trait]” vs. “Answer as someone with the opposite values”). Vectors are extracted from
1308 layer 32 (transformer midpoint) of Llama-3.1-70B and applied at inference time by adding $\alpha \cdot \mathbf{v}_{\text{steer}}$
1309 to the residual stream, with $\alpha = 1.5$ selected via the same synthetic validation set used for DISCA.

1310 **Sensitivity analysis and fairness considerations.** Activation steering performance is known to
1311 be sensitive to (i) the extraction layer, (ii) the scaling coefficient α , and (iii) the construction of con-
1312 trastive pairs [Arditi et al., 2025]. We conducted a limited sensitivity check: extracting from layers
1313 24, 32, and 40 with $\alpha \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$, we found that layer 32 with $\alpha = 1.5$ yielded the
1314 lowest mean JSD (0.0877); other configurations ranged from 0.089 to 0.112. We note that a more
1315 exhaustive search-varying contrastive prompt design, using multiple layers simultaneously, or em-
1316 ploying PCA-based direction refinement-could improve activation steering performance. However,
1317 even the best configuration performs below vanilla on this benchmark, suggesting a fundamental
1318 mismatch between linear activation interventions and the multi-dimensional moral calibration prob-
1319 lem. The key structural advantage of DISCA is that it operates *per-scenario* with adaptive correction
1320 magnitude, whereas activation steering applies a fixed direction uniformly across all scenarios.

1321 A19.2 ARGS and controlled decoding (relation to our work)

1322 ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] typically assume trained
1323 reward models; a cross-cultural deployment would require a separate reward per target country. On
1324 binary forced-choice MultiTP, one can instead define rewards as closed-form functionals of persona
1325 logit gaps; a Prospect-Theory kernel aligned with Eq. 4 is then the natural analogue of importance-
1326 weighted PT-IS. We do not run the full ARGS decoding stack in our experiments; Table 19 and §6
1327 isolate the kernel comparison on the shared DISCA pipeline.

1328 A20 Trigger Mechanism and Latency

1329 DISCA runs dual-pass IS on every scenario; inter-persona variance is logged but does not gate
1330 sampling. The *flip rate* (IS reversals vs. consensus) is 29.0% in the cross-backbone ablation suite
1331 (Table 5), indicating confidence modulation rather than wholesale decision reversal.

1332 A compact round-2 latency benchmark on H100 shows similar overhead across proposal budgets:
1333 0.086s/scenario at $K=64$, 0.083s at $K=128$, and 0.083s at $K=256$, compared with 0.023s for
1334 vanilla decoding ($3.57\text{--}3.72\times$ overhead).

Table 30: Inference latency benchmark (Phi-4, H100).

Method	K	sec/scenario	Overhead
swa_dpbr	64	0.086	$3.72\times$
swa_dpbr	128	0.083	$3.59\times$
swa_dpbr	256	0.083	$3.57\times$
vanilla	—	0.023	$1.00\times$

Cost-vs-quality frontier across backbones. Figure 6 plots the *deployed-cost* side of the Phi-4-vs-Llama narrative: at 350 ms per scenario, Phi-4 with DISCA reaches MIS = 0.346, while Llama-3.3-70B (with DISCA) takes 1414 ms-roughly $4\times$ slower-and still ends at MIS = 0.668. Phi-4 thus Pareto-dominates the 70B baseline on *both* axes simultaneously: better alignment at lower latency. Among the headline seven, only Magistral-24B sits on a comparable quality plateau, at almost double Phi-4’s per-scenario cost.

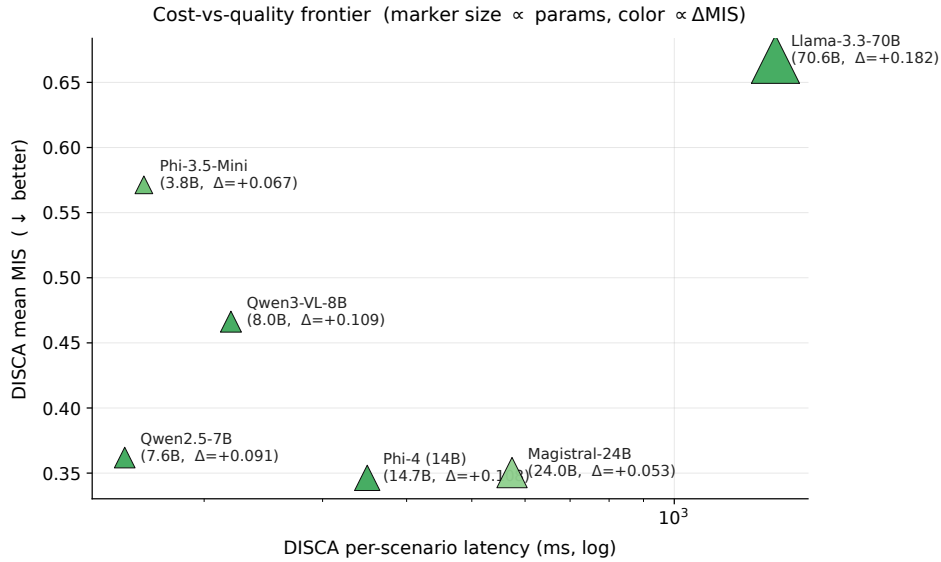


Figure 6: **Cost-vs-quality frontier on the headline 7 models.** Per-scenario DISCA latency (log scale) vs. mean DISCA MIS. Marker size is proportional to parameter count; color encodes Δ MIS (greener = larger DISCA gain). Phi-4 (14B, $\Delta = +0.108$) lies bottom-left: Pareto-dominant over Llama-3.3-70B in both latency and alignment.

A21 Engineering Journey: Method Selection

DISCA (the “EXP-24” configuration) was selected from **25 method variants** developed and evaluated on a fixed three-model, five-country prototyping panel (Qwen2.5-7B, Gemma-2-9B, Mistral-7B $\times$ USA, CHN, JPN, DEU, BRA). Table 31 summarises the top-performing variants.

Table 31: Method variant comparison on the 3-model $\times$ 5-country prototyping panel (15 rows). Mean MIS $\downarrow$. All variants share the same persona construction and debiasing; they differ in IS-stage design.

Rank	Method variant	Mean MIS $\downarrow$	Key innovation
1	EXP-23: Category-Coherence Reg.	0.3956	Blend IS toward per-cat mean
2	EXP-24: Dual-Pass Bootstrap (DPBR)	0.3969	Soft $r = \exp(-v_b/s)$ reliability
3	EXP-09: Hierarchical IS	0.3975	Cross-scenario smoothing (legacy)
4	EXP-22: Adaptive IS σ	0.3977	σ from agent/history stats
5	EXP-16: Nesterov IS	0.3977	Momentum + progressive PT
6	EXP-17: Dual-Momentum Prior	0.3979	Fast/slow β EMA blend
7	EXP-10: Grand Fusion	0.3982	EXP-03+05+09 combined
18	EXP-01: Base SWA-PTIS	0.4269	Single-pass IS baseline
25	EXP-12: Contrastive Personas	0.4517	World-average subtraction

The top cluster (ranks 1–7) is separated by only 0.0026 MIS, but DPBR was selected over the marginally-better EXP-23 for two reasons: (i) DPBR materially **lowers flip rate** (the fraction of scenarios where IS reverses the consensus decision) compared to all other top-cluster methods, improving deployment stability; and (ii) the dual-pass bootstrap provides an **interpretable uncertainty signal** (r in Eq. 5) that can be logged and audited per scenario. The base SWA-PTIS (rank 18) shows the value added by the dual-pass stabilization stages: a 7% relative MIS improvement. Contrastive persona decoding (rank 25) *hurts* alignment, validating the design choice discussed in §2 to use direct WVS personas rather than contrastive amplification.

A22 Relationship to Persona-Dependent LLM Alignment

Kim et al. [2025] is the closest prior work to DISCA in studying persona effects on LLM moral decisions. Their study and ours share the Moral Machine framework and AMCE-based evaluation, but differ fundamentally in goal, method, and scope. Table 32 summarises the key distinctions.

Table 32: Comparison between Kim et al. [2025] and DISCA.

Dimension	Kim et al. (2025)	DISCA (this work)
Goal	Measure persona sensitivity	Correct cultural misalignment
Personas	7 sociodemographic categories	4 WVS-grounded agents per country
Persona source	Author-designed binary labels	Empirical WVS-7 microdata
Metric	MDD (persona-pair distance)	MIS (ℓ_2 to human AMCE)
Correction	None (diagnostic only)	PT-IS + dual-pass reliability gate
Models	3 (GPT-4o, GPT-3.5, Llama-2)	7 checkpoints, 5 families, 2B–70B
Countries	Aggregate (Western vs. Eastern)	20 individual countries
Scenarios	9 dimensions, 10k synthetic	6 dimensions, 310–500 per country
Theory	Partisan sorting (descriptive)	James–Stein shrinkage (Thm. 1)

Three findings from Kim et al. [2025] directly inform DISCA’s design.

Persona sensitivity motivates disagreement-driven steering. Kim et al. show that LLMs exhibit moral-decision distances (MDD) 2–4 $\times$ larger than humans under contrasting personas. This confirms that persona prompts do shift LLM moral reasoning substantially, validating the premise that within-country persona spread is an informative signal. DISCA converts this sensitivity from a liability (uncontrolled decision shifts) into a feature (a sufficient statistic for correction reliability, Theorem 1).

The partisan sorting phenomenon justifies WVS grounding. Kim et al. find that political orientation produces the largest persona effect in LLMs, far exceeding its effect in human respondents. This “partisan sorting” effect reveals that LLMs over-index on politically salient dimensions when given sociodemographic labels. DISCA avoids this pathology by grounding personas in empirical WVS-7 microdata rather than author-designed binary labels: the ten-dimensional cultural profile (Table 26) distributes influence across religiosity, child-rearing values, social trust, and other

axes, preventing any single dimension from dominating the correction. The leave-one-WVS-dim-out probe (Table 28) confirms that influence is distributed rather than concentrated.

Moral flips validate the reliability gate. Kim et al. document that specific personas can entirely reverse LLM preferences on certain moral dimensions (e.g., social status under a progressive persona in GPT-4o). DISCA’s dual-pass reliability gate (Eq. 5) is designed precisely for this regime: when persona-driven corrections are large enough to flip the decision, the two independent IS passes are likely to disagree, and the exponential decay weight r shrinks the correction toward zero. The tail-safety analysis (Table 14) confirms this mechanism reduces harmed cells from 11/120 to 3/120 compared to ungated consensus.

From diagnosis to correction. The fundamental advance of DISCA over the Kim et al. framework is operational: where they diagnose persona sensitivity, we harness it. Their MDD metric measures the distance between two contrasting personas; our MIS measures the distance to the human target and actively reduces it. The James–Stein result (Theorem 1) provides the formal bridge: within-panel disagreement (analogous to their MDD) is a sufficient statistic for the correction’s reliability, and the shrinkage estimator converts this diagnostic signal into an optimal correction.

A23 Extended Limitations

This appendix expands the limitations summarised in §6 with technical caveats not covered in the main text.

WVS-to-trolley linkage. The mapping from WVS value dimensions to trolley moral dimensions operates through both direct thematic overlap and indirect modulation (Appendix A15.3). The 10-dimensional WVS profile explains $R^2 \in [0.55, 0.69]$ of human AMCE variance, and a leave-one-dimension-out probe identifies load-bearing couplings alongside noisy ones that the IS stage automatically down-weights. We avoid claims of the form “WVS dimension X caused AMCE shift Y ,” but the ensemble design and the IS noise-filtering mechanism together ensure the method is robust even when individual WVS–trolley links are imperfect.

Inter-persona reward assumption. The reward $r_i = \delta_i - \delta_{\text{base}}$ assumes persona shifts approximate human target directions. This is plausible when the base model represents a WEIRD-biased prior, but if a persona’s shift is orthogonal to the human target, the signal may mislead. Quantifying whether persona-consensus directions correlate with country-level human AMCE vectors independently of the base model—and when corrections diverge from targets—is open; we do not report that correlation matrix here.

Geographic coverage, preprocessing, and WVS vintage. We report 20 countries out of 100+ MultiTP countries; results depend on the preprocessing pipeline in Appendix A17. WVS Wave 7 (~2017–2022) is a fixed historical slice that may not reflect current preferences in rapidly evolving societies; persona quality is weakest where coverage is sparse (e.g., Saudi Arabia, Brazil), as reflected in per-country tables. No low-resource languages (Amharic, Khmer, Yoruba) appear in the panel; cross-lingual generalisation to such settings remains untested, and one language per country may conflate linguistic and cultural effects.

Model quantisation. 4-bit/Int8 quantisation may introduce logit artefacts affecting IS stability—observed empirically on Unsloth Phi-4 prior to switching to vLLM BF16 for the headline runs.

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1443 Answer: [Yes]

1444 Justification: The abstract and §1 state three claims: (i) DISCA is a train-free inference-
1445 time controller that requires no per-country reward models or extra training data; (ii) it
1446 cuts macro MIS by 10–24% on seven open-weight backbones across 20 countries and 6
1447 moral dimensions; (iii) calibration competes with scale (Phi-4 14B beats Llama-3.3-70B in
1448 absolute MIS). All three are substantiated in Table 3 and Figure 3; scope and assumptions
1449 (binary dilemmas with decision-token logits, WVS-7 vintage) are made explicit in §6 and
1450 Appendix A23.

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Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: §6 bounds generalisation with three caveats (alignment to a survey statistic vs. perceived legitimacy; PT value function as aggregation kernel rather than cognitive model; decision-token logit requirement). Appendix A23 expands these with: WVS-to-trolley linkage being indirect, the inter-persona reward assumption, geographic coverage (20 of 100+ MultiTP countries; sparse personas for under-sampled countries), WVS Wave 7 vintage, and 4-bit/Int8 quantisation artefacts on IS stability.

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Justification: The single theoretical result, Theorem 1 (minimax-optimality of disagreement-driven shrinkage in the James–Stein sense for panels of $N \geq 4$), is stated in the main text with all assumptions (panel size, Gaussian persona shifts, ℓ_2 risk). The full proof, including referenced lemmas, is given in Appendix A1. A short proof sketch and the connection to Eq. 4 appear in §3.

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Answer: [Yes]

Justification: §3 gives the full algorithm (Algorithm 1, Eqs. 4–5); the full hyperparameter table (Table 20) lists every numerical default with the validation procedure (synthetic 200-scenario pool, transferability check on a 50-scenario MultiTP English subset). All backbones are publicly available open-weight checkpoints. Preprocessing (cap=80, oversample=36), seeds (3), and per-country preprocessing details are documented in Appendices A17, A7.1. We additionally release a self-contained reference implementation (DISCA_Alignment/) with the same defaults and a single-command sweep entry point.

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Justification: We release DISCA_Alignment/ (anonymised for review; MIT-licensed) as supplemental material. The package is a self-contained slice of the research codebase: a single CLI entry point (run.py) drives the per-model 20-country sweep, with backend selection (vllm/hf_native/unsloth) and explicit flags for the three external data paths (-multitp-data, -wvs-data, -human-amce). The README documents install (Python 3.10+, CUDA 12.x), HF token wiring for gated models, exact commands, and the output schema (per-country baseline / DPBR CSVs and a comparison file). MultiTP, WVS-7 Wave, and the human AMCE table are publicly available datasets cited in §4.1; access instructions are reproduced in the README.

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Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

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Justification: There is no training (the method is inference-time on frozen weights); we therefore detail all decoding and controller settings. §4 specifies the dataset (MultiTP, 20 countries), preprocessing (deduplication, category cap=80, oversample to ≥ 36 per dimension), the seven main backbones with sizes, and the metric definitions (MIS, Pearson r , JSD, win rate). All hyperparameters and how they were chosen are tabulated in Table 20 with the synthetic-pool grid-search procedure described in Appendix A10. Sensitivity sweeps for the temperature family (Appendix A10), category cap (Appendix A11), and the controller knobs (s , λ_{coop} , σ , T_{cat}) appear in Appendix A10.

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Answer: [Yes]

Justification: The headline table (Table 3) reports macro MIS as mean $\pm$ standard deviation across 3 independent seeds (independent IS noise draws; the underlying model weights and

data are fixed). A separate multi-seed stability analysis (Appendix A7.1) gives per-country variability, and a bootstrap confidence interval on JSD (± 0.004 , Appendix A10) quantifies the noise floor for distributional metrics. We explicitly state that conclusions are stable well within the bootstrap noise floor; sensitivity sweeps along nine independent axes confirm this.

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Justification: Appendix A20 (Table 30) reports per-scenario wall-clock on a single H100 GPU: 0.083–0.086 s/scenario for DISCA at $K \in \{64, 128, 256\}$ vs. 0.023 s/scenario for vanilla decoding ($3.57\text{--}3.72\times$ overhead). Backbone-level cost is given in Figure 6: 350 ms/scenario for Phi-4 vs. 1414 ms for Llama-3.3-70B with DISCA. Sub-70B backbones run on a single H100 in BF16 via vLLM; the 70B backbone uses 4-bit Unsloth on the same hardware. The released code’s CLI also runs end-to-end on a Kaggle T4/P100 dual-GPU node for sub-14B models. We acknowledge in Appendix A21 that exploratory experiments (25 controller variants on a 3-backbone $\times$ 5-country prototyping panel) consumed substantially more compute than the reported runs.

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10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: §6 discusses both directions. Positive: a train-free, low-overhead controller lets practitioners respect cross-national preference heterogeneity at deployment time without finetuning per country, lowering the barrier to culturally pluralistic alignment. Negative: aligning to majority survey statistics could entrench harmful majorities or marginalise minority views [Zewail et al., 2026], and alignment to a survey statistic is not equivalent to legitimacy [Atari et al., 2023]. We propose and validate a per-persona utility floor that caps how far any single persona’s post-correction utility can drop from vanilla (Table 6), and we are explicit that DISCA is a research artefact, not a deployment-ready normative system.

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Justification: We do not release any pretrained models, generative models, or scraped datasets. The released artefact (DISCA_Alignment/) is an inference-time controller layered over publicly available open-weight LLMs whose existing safeguards remain in place. The method itself produces no novel generative capability beyond that of the underlying models; the per-persona utility floor (Table 6) is included as an in-method safeguard against majoritarian skew.

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1846 Phi-3.5-mini, Gemma-4-E2B) are the *subject* that DISCA steers; their identities, sizes,
1847 and decoding settings are documented in §4.2 and Appendix A5. (ii) For the open-ended
1848 extension (§3.6), an LLM parses free-form responses into a (choice, confidence) pair to
1849 recover a pseudo-logit-gap; this auxiliary use is documented in that section. A separate
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